

Bayesian Inference

A Continuous Treatment of Lectures 3–12

From the Beta–Bernoulli Model to Multi-Parameter Normal Inference,
Jeffreys Priors, and Credible Intervals

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- Part I** Foundations: The Three Steps, Exchangeability, Beta–Bernoulli (Lec. 3–4)
 - Part II** Conjugate Priors: Beta–Bernoulli and Normal–Normal (Lec. 5–6)
 - Part III** Improper Priors, Jeffreys Priors, Poisson Regression (Lec. 7–8)
 - Part IV** Fisher Information and Multi-Parameter Models (Lec. 9–10)
 - Part V** The Full Normal Model, Reference Priors, MLE/MAP, Credible Intervals (Lec. 11–12)

Compiled and unified from handwritten lecture notes, with full derivations,
worked examples, cross-referencing tables, and connective commentary.

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Preface: How This Book Is Organized

This book compiles five sets of lecture notes – covering Lectures 3 through 12 of a course on Bayesian inference – into a single, continuous narrative. The original notes were taken across five separate sessions, each with its own notation habits and its own starting point. Read independently, they are five islands. The purpose of this book is to build the bridges: to show that Lecture 3’s “three steps of Bayesian analysis” are the scaffolding that every later chapter fills in; that the Beta–Bernoulli model of Lecture 4 is the same Beta–Bernoulli model re-derived more carefully in Lecture 5; that the Jeffreys prior introduced in Lecture 7 for a single parameter is the exact same object re-derived, generalized, and applied in Lectures 9–12; and that the credible intervals of the final lecture are the natural destination of everything that came before.

A single running thread. Every chapter in this book returns to the same underlying question: *given a prior $\pi(\theta)$ and a likelihood $f(x | \theta)$, what is the posterior $\pi(\theta | x)$, and what do we do with it?* The models change – Bernoulli, Normal, Poisson – and the priors change – flat, conjugate, Jeffreys – but the mechanics never do:

$$\pi(\theta | x) \propto \pi(\theta) f(x | \theta).$$

Keep this equation in view. Nearly every derivation in this book is an exercise in matching the right-hand side to a recognizable kernel.

How to use the “Key Idea,” “Recipe,” and “Connecting the Threads” boxes. Throughout the book, blue *Key Idea* boxes isolate the single sentence you should remember from a section. Recipe boxes give step-by-step procedures you can reuse on a new problem. Red *Connecting the Threads* boxes are new to this compilation – they did not appear in the original notes – and exist specifically to tie a new result back to something already proved, or forward to something still to come.

On the exercises. The original notes repeatedly flagged calculations as “to derive by hand” or left as board exercises. Every one of these has been carried out in full in this book, so that the text is self-contained; where the original notes intended active practice, a parallel *Exercise* box is retained so the reader can still test themselves before reading the solution that follows.

Roadmap of the Course

Lectures	Model / Tool	What gets established
3–4	Beta–Bernoulli, exchangeability	The three steps of Bayesian analysis; posterior odds; exchangeability and de Finetti’s theorem as the justification for “i.i.d. given θ ”; first posterior derivation under a flat prior.
5–6	Beta–Bernoulli (general); Normal–Normal	Full conjugacy proofs for general Beta priors; posterior mean as a shrinkage estimator; posterior variance and consistency; Normal–Normal conjugacy for one and n observations; the multivariate normal (MVN) route to the same results.
7–8	Improper priors; Jeffreys priors; Poisson regression	When a prior need not integrate to 1; propriety of the posterior; Fisher information; the Jeffreys prior and its invariance; first derivations for Bernoulli and Poisson; KL divergence as curvature; Poisson regression with covariates.
9–10	Fisher information; multi-parameter models	Cramér–Rao bound; KL divergence curvature proved rigorously; full table of Jeffreys priors; nuisance parameters and marginalization; Monte Carlo marginalization; Gibbs sampling; the scaled inverse- χ^2 distribution.
11–12	Full Normal model; reference priors; MLE/MAP; credible intervals	Joint posterior of (μ, σ^2) under the independence Jeffreys prior; pivotal-quantity derivation of reference priors (an independent route to the same priors as Jeffreys’ rule); invariance of the MLE versus non-invariance of the MAP; credible intervals and their contrast with confidence intervals.

Part I

Foundations: The Three Steps, Exchangeability, and the Beta–Bernoulli Model

1 The Anatomy of a Bayesian Analysis

1.1 The Three Steps

Every Bayesian analysis, however elaborate it eventually becomes, is built from the same three moves. Stating them clearly at the outset is worth the trouble, because every later chapter of this book is simply a worked instance of these three steps applied to a specific model.

Step 1 – Set up a full probability model. Construct a joint probability distribution for *all* observable and unobservable quantities, consistent with what is known about the underlying scientific problem and the data-collection process.

Step 2 – Condition on the observed data. The posterior distribution is the conditional distribution of the unobserved quantities, given the observed data.

Step 3 – Evaluate the fit of the model. Once a posterior has been obtained, check whether the model is a reasonable description of the data (e.g. via posterior predictive checks).

A running example. Take n coin tosses with unknown bias θ . The *unobservable* quantity is θ ; the *observables* are $y_1, \dots, y_n$. Step 1 asks us to write down the joint distribution of everything:

$$p(\theta, y_1, \dots, y_n) = p(\theta) p(y_1, \dots, y_n \mid \theta).$$

This factorization – prior times likelihood – is the single most important equation in the book; every model considered from here on (Bernoulli, Normal, Poisson) is just a different choice of $p(\theta)$ and $p(y_1, \dots, y_n \mid \theta)$ plugged into the same factorization. Step 2 is the algebra of turning this joint distribution into $p(\theta \mid y_1, \dots, y_n)$ via Bayes’ rule – the subject of essentially every chapter that follows. Step 3, model checking, is flagged in the original notes as a topic for later in the course and is not developed further here.

Key Idea

Steps 1 and 2 are two sides of the same joint distribution: Step 1 builds it, Step 2 slices it. Nothing new needs to be invented for Step 2 beyond ordinary conditional probability.

1.2 Posterior Odds versus Prior Odds

A common way to use a posterior is not to summarize it by a mean or a mode, but to compare two competing values of θ – for instance, to ask whether the data favor θ_1 (e.g., “the estimator is unbiased”) or θ_2 (“it is biased”). Define

$$\text{prior odds} = \frac{p(\theta_1)}{p(\theta_2)}, \quad \text{posterior odds} = \frac{p(\theta_1 \mid y)}{p(\theta_2 \mid y)}.$$

Proposition 1.1 (Posterior odds decompose into prior odds times a likelihood ratio).

$$\frac{p(\theta_1 \mid y)}{p(\theta_2 \mid y)} = \underbrace{\frac{p(\theta_1)}{p(\theta_2)}}_{\text{prior odds}} \times \underbrace{\frac{p(y \mid \theta_1)}{p(y \mid \theta_2)}}_{\text{likelihood ratio (Bayes factor)}}.$$

Proof. Apply Bayes’ rule to each hypothesis separately and divide; the common denominator $p(y)$ cancels:

$$\frac{p(\theta_1 \mid y)}{p(\theta_2 \mid y)} = \frac{p(\theta_1)p(y \mid \theta_1)/p(y)}{p(\theta_2)p(y \mid \theta_2)/p(y)} = \frac{p(\theta_1)}{p(\theta_2)} \cdot \frac{p(y \mid \theta_1)}{p(y \mid \theta_2)}.$$

□

Consequently

$$\frac{p(\theta_1 | y)}{p(\theta_2 | y)} \geq \frac{p(\theta_1)}{p(\theta_2)} \iff \frac{p(y | \theta_1)}{p(y | \theta_2)} \geq 1 :$$

the posterior odds move away from the prior odds in whichever direction the likelihood ratio points. If the data are more likely under θ_1 , belief shifts toward θ_1 , and vice versa.

Theorem 1.2 (Consistency of posterior odds). *Suppose $y_1, y_2, \dots$ are drawn i.i.d. from the true model $f(\cdot | \theta^*)$ with $\theta^* \in \{\theta_1, \theta_2\}$. Then, almost surely,*

$$\frac{p(y_1, \dots, y_n | \theta_1)}{p(y_1, \dots, y_n | \theta_2)} \longrightarrow \begin{cases} \infty & \text{if } \theta^* = \theta_1, \\ 0 & \text{if } \theta^* = \theta_2. \end{cases}$$

Proof. Write $L_n := \log \frac{p(y_1, \dots, y_n | \theta_1)}{p(y_1, \dots, y_n | \theta_2)} = \sum_{i=1}^n \log \frac{f(y_i | \theta_1)}{f(y_i | \theta_2)}$, a sum of i.i.d. terms. Suppose $\theta^* = \theta_1$. By the strong law of large numbers,

$$\frac{1}{n} L_n \xrightarrow{\text{a.s.}} \mathbb{E}_{\theta_1} \left[\log \frac{f(Y | \theta_1)}{f(Y | \theta_2)} \right] = D_{\text{KL}}(f(\cdot | \theta_1) \| f(\cdot | \theta_2)) > 0,$$

where the inequality is Gibbs' inequality (the Kullback–Leibler divergence between two distinct distributions is strictly positive). Since L_n/n converges a.s. to a strictly positive constant, $L_n \rightarrow \infty$ a.s., so the likelihood ratio $e^{L_n} \rightarrow \infty$. The case $\theta^* = \theta_2$ is symmetric, giving $L_n \rightarrow -\infty$ and the ratio $\rightarrow 0$. $\square$

Key Idea

This is why posterior odds are a self-correcting procedure: as $n \rightarrow \infty$, the likelihood ratio dominates whatever prior odds you started with, and the posterior concentrates on whichever hypothesis actually generated the data. This same mechanism – the log-likelihood ratio behaving like a random walk with a drift set by a KL divergence – reappears in Chapter 6 as the local curvature that defines Fisher information.

1.3 Conditional Independence and Exchangeability

Consider ten people's measurements $y_1, \dots, y_{10}$ (say, heights in cm), and ask how to predict an eleventh person's measurement y_{11} .

Conditional independence given θ . A new observation is independent of past observations *once θ is known*:

$$p(y_{n+1} | \theta, y_1, \dots, y_n) = p(y_{n+1} | \theta).$$

But marginally (before conditioning on θ),

$$p(y_{n+1} | y_1, \dots, y_n) = f(y_1, \dots, y_n)$$

depends on the past data – because the past data inform our beliefs about θ , which in turn inform the prediction of y_{n+1} . This is the basic mechanism of learning from data in the Bayesian framework: information never flows directly between observations, only through the parameter.

Definition 1.3 (Exchangeability). $y_1, \dots, y_n$ are *exchangeable* if their joint distribution is invariant under any permutation π of the indices:

$$p(y_1, \dots, y_n) = p(y_{\pi(1)}, \dots, y_{\pi(n)}) \quad \text{for every permutation } \pi.$$

Key Idea

Exchangeability is a *weaker and more natural* assumption to justify directly from modeling considerations than “i.i.d.” – it only asks that the labeling of observations carries no information, not that they are literally generated by independent identical mechanisms. Section 1.4 shows that, remarkably, exchangeability is not really weaker in its consequences: an infinitely exchangeable sequence is *equivalent* to the existence of a parameter θ and prior $p(\theta)$ under which the sequence is i.i.d. given θ .

Why exchangeable does not imply i.i.d. unconditionally. It is easy to see that i.i.d. implies exchangeable: permuting a product of identical, independent factors does not change the product. The converse fails. Consider drawing two balls without replacement from an urn containing one red and one blue ball, and let $Y_i = 1$ if the i -th draw is red. Then $(Y_1, Y_2) \in \{(1, 0), (0, 1)\}$, each with probability $\frac{1}{2}$, so $p(Y_1, Y_2) = p(Y_2, Y_1)$ and the pair is exchangeable – but Y_1 and Y_2 are certainly not independent (knowing Y_1 tells you Y_2 exactly), so the sequence is not i.i.d. unconditionally.

1.4 De Finetti’s Representation Theorem

Theorem 1.4 (De Finetti). *If $Y_1, Y_2, \dots$ is an infinitely extendable exchangeable sequence of binary outcomes, then there exists a distribution $p(\theta)$ such that, for every n ,*

$$p(y_1, \dots, y_n) = \int_0^1 \theta^{\sum_i y_i} (1 - \theta)^{n - \sum_i y_i} p(\theta) d\theta.$$

Equivalently: $Y_1, \dots, Y_n \mid \theta \stackrel{iid}{\sim} \text{Bernoulli}(\theta)$ for some random $\theta \sim p(\theta)$, and $\sum_{i=1}^n Y_i \mid \theta \sim \text{Binomial}(n, \theta)$.

The hypothesis of *infinite* exchangeability – that the sequence can be extended to arbitrarily many further exchangeable observations, not just that some fixed $y_1, \dots, y_n$ happen to be exchangeable – is essential; finite exchangeability alone does not guarantee the mixture representation holds exactly (the urn example above is finitely exchangeable for $n = 2$ but obviously cannot be extended to an infinite i.i.d.-given- θ mixture, since only two draws are even possible).

Connecting the Threads

De Finetti’s theorem is the theoretical justification for something this book will do without comment starting in the very next section: write down a likelihood $Y_i \mid \theta \stackrel{iid}{\sim} \text{Bernoulli}(\theta)$ and a prior $p(\theta)$, and treat that as an unproblematic modeling choice. Whenever you are willing to assume only exchangeability of your data – a much weaker and more defensible assumption than “i.i.d.” – de Finetti guarantees that *some* parameter θ and prior $p(\theta)$ exist making the i.i.d.-given- θ picture exact. This is the conceptual bridge connecting Step 1 of Chapter 1 (“construct a joint distribution”) to every conjugate model in Parts II–V.

2 The Beta–Bernoulli Model: First Pass

2.1 Setting Up the Model

With the machinery of Chapter 1 in hand, we now carry out a first complete Bayesian analysis. Let

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta), \quad y = (y_1, \dots, y_n),$$

and write $z = \sum_i y_i$ for the number of successes. The three ingredients are

$$\underbrace{p(\theta)}_{\text{prior}}, \quad \underbrace{p(y \mid \theta) = \theta^z (1 - \theta)^{n-z}}_{\text{likelihood}}, \quad \underbrace{p(\theta \mid y)}_{\text{posterior}} \propto p(\theta) p(y \mid \theta).$$

A random variable $X \sim \text{Beta}(a, b)$ has density $f_X(x) \propto x^{a-1}(1-x)^{b-1}$ on $(0, 1)$.

2.2 The Uniform Prior Special Case

Take $\theta \sim \text{Uniform}(0, 1)$, i.e. $p(\theta) = 1$. Then

$$p(\theta \mid y) \propto \theta^z (1 - \theta)^{n-z} \implies \theta \mid y \sim \text{Beta}(z + 1, n - z + 1).$$

This is the pattern-matching step referenced in the Preface: the un-normalized posterior is literally the kernel of a Beta density, so no integration is required to identify it.

Proposition 2.1 (Marginal distribution of z under the uniform prior). *With $p(\theta) = 1$, the prior predictive distribution of $z = \sum_i Y_i$ is discrete uniform on $\{0, 1, \dots, n\}$: $p(z) = \frac{1}{n+1}$ for every $z \in \{0, \dots, n\}$.*

Proof. By the law of total probability and the Beta function identity $B(z+1, n-z+1) = \frac{z!(n-z)!}{(n+1)!}$,

$$p(z) = \binom{n}{z} \int_0^1 \theta^z (1 - \theta)^{n-z} d\theta = \binom{n}{z} B(z+1, n-z+1) = \frac{n!}{z!(n-z)!} \cdot \frac{z!(n-z)!}{(n+1)!} = \frac{1}{n+1}. \quad \square$$

This is a mildly surprising fact worth pausing on: under a flat prior on θ , *every* possible count of successes, from 0 to n , is equally likely *before* the data are observed – the flatness of the prior on θ does not translate into flatness of belief about the count z in any simple way except this exact uniform-on- $\{0, \dots, n\}$ result.

2.3 Posterior Prediction and Laplace’s Rule of Succession

Proposition 2.2.

$$\Pr(\tilde{y} = 1 \mid y) = \int_0^1 \theta p(\theta \mid y) d\theta = \mathbb{E}[\theta \mid y].$$

Proof. Condition on θ first and use conditional independence (Section 1.3): $\Pr(\tilde{y} = 1 \mid \theta, y) = \theta$, so by the law of total probability, $\Pr(\tilde{y} = 1 \mid y) = \int \Pr(\tilde{y} = 1 \mid \theta, y) p(\theta \mid y) d\theta = \int \theta p(\theta \mid y) d\theta. \quad \square$

Using $\theta \mid y \sim \text{Beta}(z + 1, n - z + 1)$ and the Beta mean formula $E(X) = \frac{a}{a+b}$,

$$\Pr(\tilde{y} = 1 \mid y) = E[\theta \mid y] = \frac{z + 1}{n + 2}.$$

Key Idea

This is *Laplace’s rule of succession*. Compare it to the maximum likelihood estimate $\hat{\theta}_{\text{MLE}} = z/n$: the posterior mean shrinks the raw sample proportion slightly toward $\frac{1}{2}$, and the size of this shrinkage vanishes as $n \rightarrow \infty$. Shrinkage of exactly this flavor – a weighted compromise between a “prior guess” and the data, with the data’s weight growing with n – is the single most recurring phenomenon in this book. It reappears, in more general form, in Proposition 3.4 (general Beta prior) and Theorem 4.1 (Normal–Normal conjugacy).

2.4 Summarizing the Posterior

Quantity	Formula (for $\theta \mid y \sim \text{Beta}(z + 1, n - z + 1)$)
Posterior mean	$E(\theta \mid y) = \frac{z + 1}{n + 2}$
Posterior mode	$\text{Mode}(\theta \mid y) = \frac{z}{n}$ (coincides with $\hat{\theta}_{\text{MLE}}$)
Posterior variance	$\text{Var}(\theta \mid y) = \frac{(z + 1)(n - z + 1)}{(n + 2)^2(n + 3)}$
MLE	$\hat{\theta}_{\text{MLE}} = z/n$

Derivation of the mode and variance. For $X \sim \text{Beta}(a, b)$ with $a, b > 1$, differentiating $\log f_X(x) = (a - 1) \log x + (b - 1) \log(1 - x) + \text{const}$ and setting the derivative to zero gives $\frac{a - 1}{x} = \frac{b - 1}{1 - x}$, so $\text{Mode}(X) = \frac{a - 1}{a + b - 2}$. With $a = z + 1$, $b = n - z + 1$ this is $\frac{z}{n}$ – exactly the MLE. (Chapter 3 shows this coincidence of posterior mode and MLE under a flat prior is a special case of a much more general fact about improper/Jeffreys priors.) For the variance, $\text{Var}(X) = \frac{ab}{(a + b)^2(a + b + 1)}$ (proved in full in Lemma 3.5); substituting $a = z + 1, b = n - z + 1$ gives the stated formula. $\square$

The posterior density $\text{Beta}(z + 1, n - z + 1)$ is unimodal on $(0, 1)$: it rises from 0 at $\theta = 0$ to a single peak and falls back to 0 at $\theta = 1$, skewed toward whichever side z favors. A standard two-sided credible interval for θ reports the 0.025 and 0.975 quantiles (or 0.05/0.95 for a 90% interval) of this Beta distribution – the precise machinery for this is developed in Chapter 15.

Connecting the Threads

Everything in this chapter was done under the special case $\theta \sim \text{Uniform}(0, 1)$. The very first task of Part II is to redo this derivation for an arbitrary $\text{Beta}(\alpha, \beta)$ prior, and to prove – not just observe – that the Beta family is *conjugate* to the Bernoulli likelihood.

Part II

Conjugate Priors: Beta–Bernoulli and Normal–Normal

3 The Beta–Bernoulli Model, Completed

Chapter 2 treated only the special case of a flat prior. We now redo the analysis with a general $\text{Beta}(\alpha, \beta)$ prior, prove conjugacy formally, and extract two further structural results – the posterior mean as a Bayes estimator, and posterior consistency – that will recur, in different clothing, throughout the rest of the book.

3.1 Conjugacy for the General Beta Prior

Definition 3.1 (Conjugate prior). A family of priors $\{\pi_\eta(\theta)\}$ is *conjugate* to a likelihood $p(x \mid \theta)$ if, for every η , the resulting posterior $\pi(\theta \mid x)$ belongs to the same family (possibly with different parameters η'). Conjugacy converts Bayesian updating into simple algebra on the parameters, with no integral ever needing to be evaluated.

Theorem 3.2 (Beta–Bernoulli conjugacy). Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bernoulli}(\theta)$ and suppose $\theta \sim \text{Beta}(\alpha, \beta)$. Then

$$\theta \mid x_1, \dots, x_n \sim \text{Beta}\left(\alpha + \sum_i x_i, \beta + n - \sum_i x_i\right).$$

Proof. By Bayes' rule, $p(\theta \mid x_{1:n}) \propto p(\theta) p(x_{1:n} \mid \theta)$, and

$$p(\theta \mid x_{1:n}) \propto [\theta^{\alpha-1}(1-\theta)^{\beta-1}] \cdot [\theta^{\sum x_i}(1-\theta)^{n-\sum x_i}] = \theta^{(\alpha+\sum x_i)-1}(1-\theta)^{(\beta+n-\sum x_i)-1},$$

which is exactly the un-normalized kernel of a $\text{Beta}(\alpha + \sum x_i, \beta + n - \sum x_i)$ density. Since a density is uniquely determined (up to normalization) by its shape, the posterior is exactly this Beta distribution. $\square$

Setting $\alpha = \beta = 1$ (i.e. $\text{Beta}(1, 1) = \text{Uniform}(0, 1)$) recovers Chapter 2 exactly. Intuitively, Beta parameters are read as “prior successes +1” and “prior failures +1”: observing $\sum x_i$ successes and $n - \sum x_i$ failures simply adds these counts onto the prior's parameters.

3.2 The Posterior Mean as a Bayes Estimator

Lemma 3.3 (Mean of a Beta distribution). If $Z \sim \text{Beta}(a, b)$ then $E[Z] = \frac{a}{a+b}$.

Proof.

$$E[Z] = \int_0^1 z \cdot \frac{z^{a-1}(1-z)^{b-1}}{B(a, b)} dz = \frac{1}{B(a, b)} \int_0^1 z^a(1-z)^{b-1} dz = \frac{B(a+1, b)}{B(a, b)}.$$

Using $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ and $\Gamma(a+1) = a\Gamma(a)$,

$$\frac{B(a+1, b)}{B(a, b)} = \frac{\Gamma(a+1)\Gamma(b)/\Gamma(a+b+1)}{\Gamma(a)\Gamma(b)/\Gamma(a+b)} = \frac{a\Gamma(a)}{\Gamma(a)} \cdot \frac{\Gamma(a+b)}{\Gamma(a+b+1)} = \frac{a}{a+b}. \quad \square$$

Proposition 3.4 (Posterior mean in the Beta–Bernoulli model).

$$\hat{\theta}_{\text{Bayes}} := E[\theta \mid x_{1:n}] = \frac{\alpha + \sum_i x_i}{\alpha + \beta + n}.$$

Proof. Immediate from Theorem 3.2 and Lemma 3.3. $\square$

Key Idea

Shrinkage interpretation. Writing $\bar{x} = \frac{1}{n} \sum x_i$ (the MLE) and $\theta_0 = \frac{\alpha}{\alpha+\beta}$ (the prior mean),

$$\hat{\theta}_{\text{Bayes}} = \frac{\alpha + \beta}{\alpha + \beta + n} \theta_0 + \frac{n}{\alpha + \beta + n} \bar{x},$$

a weighted average of the prior mean and the sample mean, with weights set by the “prior sample size” $\alpha + \beta$ against the actual sample size n . As $n \rightarrow \infty$, the weight on $\bar{x}$ tends to 1 and $\hat{\theta}_{\text{Bayes}} \rightarrow \bar{x}$: the prior’s influence vanishes and the Bayes estimator becomes the sample proportion. This is exactly Laplace’s rule of succession from Chapter 2 in the special case $\alpha = \beta = 1$.

3.3 Posterior Variance and Consistency

Lemma 3.5 (Variance of a Beta distribution). *If $Z \sim \text{Beta}(a, b)$ then $\text{Var}(Z) = \frac{ab}{(a+b)^2(a+b+1)}$.*

Proof. As in Lemma 3.3, $E[Z^2] = \frac{B(a+2, b)}{B(a, b)} = \frac{a(a+1)}{(a+b)(a+b+1)}$ by the same Gamma-function manipulation. Then

$$\text{Var}(Z) = E[Z^2] - (EZ)^2 = \frac{a(a+1)}{(a+b)(a+b+1)} - \frac{a^2}{(a+b)^2}.$$

Putting over the common denominator $(a+b)^2(a+b+1)$, the numerator simplifies to $a(a+1)(a+b) - a^2(a+b+1) = a[(a+1)(a+b) - a(a+b+1)] = a[(a+b) + a - a] = ab$, so $\text{Var}(Z) = \frac{ab}{(a+b)^2(a+b+1)}$. $\square$

Proposition 3.6 (Posterior variance and consistency).

$$\text{Var}(\theta \mid x_{1:n}) = \frac{(\alpha + \sum x_i)(\beta + n - \sum x_i)}{(\alpha + \beta + n)^2(\alpha + \beta + n + 1)} \approx \frac{\bar{x}(1 - \bar{x})}{n} \xrightarrow{n \rightarrow \infty} 0.$$

Proof. The exact formula follows from Lemma 3.5 with $a = \alpha + \sum x_i$, $b = \beta + n - \sum x_i$. For large n , the fixed constants α, β are negligible relative to n , so $\text{Var}(\theta \mid x) \approx \frac{n\bar{x} \cdot n(1 - \bar{x})}{n^2 \cdot n} = \frac{\bar{x}(1 - \bar{x})}{n}$, which is $O(1/n) \rightarrow 0$ since $\bar{x}(1 - \bar{x}) \leq \frac{1}{4}$. $\square$

Key Idea

This is the Bayesian analogue of the law of large numbers / consistency of the MLE: as data accumulate, the posterior concentrates tightly around the truth. Combined with $\hat{\theta}_{\text{Bayes}} \rightarrow \bar{x} \rightarrow \theta_{\text{true}}$, the entire posterior distribution collapses to a point mass at the true parameter value. The identical phenomenon – posterior mean $\rightarrow$ MLE, posterior variance $\rightarrow 0$ – reappears for the Normal–Normal model in the very next section, and again, more abstractly, in Section 14.1’s discussion of the Bernstein–von Mises theorem.

4 The Normal–Normal Model

We now move from a discrete parameter $\theta \in (0, 1)$ to a continuous parameter $\theta \in \mathbb{R}$ – typically an unknown population mean – observed through Gaussian noise of known variance σ^2 . The entire chapter is a single running application of one algebraic trick, *completing the square*, applied four times with increasing generality.

4.1 A Single Observation

$$y \mid \theta \sim N(\theta, \sigma^2), \quad \theta \sim N(\mu_0, \tau_0^2),$$

where σ^2 (data variance) and μ_0, τ_0^2 (prior mean/variance) are known constants.

Theorem 4.1 (Normal–Normal conjugacy, single observation).

$$\theta \mid y \sim N(\mu_1, \tau_1^2), \quad \mu_1 = \frac{\frac{1}{\tau_0^2}\mu_0 + \frac{1}{\sigma^2}y}{\frac{1}{\tau_0^2} + \frac{1}{\sigma^2}}, \quad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2}.$$

Proof (completing the square). By Bayes' rule, $p(\theta \mid y) \propto \exp\{-\frac{1}{2\tau_0^2}(\theta - \mu_0)^2\} \exp\{-\frac{1}{2\sigma^2}(y - \theta)^2\}$. Collect the exponent:

$$-2 \log p(\theta \mid y) \stackrel{+C}{=} \theta^2 \left(\frac{1}{\tau_0^2} + \frac{1}{\sigma^2} \right) - 2\theta \left(\frac{\mu_0}{\tau_0^2} + \frac{y}{\sigma^2} \right),$$

where $\stackrel{+C}{=}$ means “equal up to an additive constant not depending on θ .” Writing $A := \frac{1}{\tau_0^2} + \frac{1}{\sigma^2}$ and $B := \frac{\mu_0}{\tau_0^2} + \frac{y}{\sigma^2}$,

$$-2 \log p(\theta \mid y) \stackrel{+C}{=} A\theta^2 - 2B\theta = A\left(\theta - \frac{B}{A}\right)^2 - \frac{B^2}{A},$$

and $-B^2/A$ is itself a θ -free constant that can be absorbed into C . Hence $p(\theta \mid y) \propto \exp\{-\frac{A}{2}(\theta - B/A)^2\}$, the kernel of a $N(B/A, 1/A)$ density. By uniqueness of densities, $\theta \mid y \sim N(B/A, 1/A)$, i.e. $\mu_1 = B/A$ and $\tau_1^2 = 1/A$ as claimed. $\square$

Key Idea

Precision-weighting. Define *precision* as the reciprocal of variance. Then

$$\text{posterior precision} = \text{prior precision} + \text{data precision},$$

and the posterior mean is the precision-weighted average of the prior mean and the datum. This is the continuous-parameter mirror of the shrinkage formula in Proposition 3.4: there, “precision” was implicitly the Beta parameters $\alpha + \beta$ versus n ; here it is literally $1/\tau_0^2$ versus $1/\sigma^2$. If the prior is very confident (τ_0^2 small), $\mu_1 \approx \mu_0$; if the data are very informative (σ^2 small), $\mu_1 \approx y$.

4.2 n I.I.D. Observations

Theorem 4.2 (Normal–Normal conjugacy, n observations). *Let $y_1, \dots, y_n \stackrel{iid}{\sim} N(\theta, \sigma^2)$ with $\theta \sim N(\mu_0, \tau_0^2)$. Then*

$$\theta \mid y_1, \dots, y_n \sim N(\mu_1, \tau_1^2), \quad \mu_1 = \frac{\frac{1}{\tau_0^2}\mu_0 + \frac{n}{\sigma^2}\bar{y}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}, \quad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2},$$

where $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$.

Proof. Since the y_i are conditionally i.i.d. given θ , using the identity

$$\sum_{i=1}^n (y_i - \theta)^2 = \underbrace{\sum_i (y_i - \bar{y})^2}_{\text{const. in } \theta} + n(\bar{y} - \theta)^2$$

(proved by expanding $(y_i - \theta) = (y_i - \bar{y}) + (\bar{y} - \theta)$ and noting the cross term vanishes since $\sum_i (y_i - \bar{y}) = 0$), the likelihood has exactly the θ -dependence of a *single* observation $\bar{y} \sim N(\theta, \sigma^2/n)$. Applying Theorem 4.1 with y replaced by $\bar{y}$ and σ^2 replaced by σ^2/n gives the stated formulas. $\square$

Key Idea

Sufficiency. The posterior depends on the full sample only through $\bar{y}$: this is the formal meaning of $\bar{y}$ being a *sufficient statistic* for θ . As $n \rightarrow \infty$, $n/\sigma^2 \rightarrow \infty$ dominates $1/\tau_0^2$, so $\mu_1 \rightarrow \bar{y}$ and $\tau_1^2 \rightarrow \sigma^2/n \rightarrow 0$ – the continuous-parameter analogue of the Beta–Bernoulli consistency result of Chapter 3.

4.3 The Marginal (Prior Predictive) Distribution of y

Before any data are observed, what is the marginal distribution of y , having integrated out uncertainty in θ ? This *prior predictive distribution* matters for model checking (Step 3 of Chapter 1) and for the multivariate-normal viewpoint below.

Theorem 4.3. If $\theta \sim N(\mu_0, \tau_0^2)$ and $y | \theta \sim N(\theta, \sigma^2)$, then marginally $y \sim N(\mu_0, \tau_0^2 + \sigma^2)$.

Proof (via sums of independent normals – the clean route). Write $y = \theta + \varepsilon$ with $\theta \sim N(\mu_0, \tau_0^2)$, $\varepsilon \sim N(0, \sigma^2)$, $\theta \perp \varepsilon$ – this is just a restatement of $y | \theta \sim N(\theta, \sigma^2)$, with $\varepsilon := y - \theta$ the independent observation noise. If $U \sim N(m_1, s_1^2)$ and $V \sim N(m_2, s_2^2)$ are independent, their moment generating functions multiply: $M_{U+V}(t) = M_U(t)M_V(t) = \exp\{(m_1 + m_2)t + \frac{1}{2}(s_1^2 + s_2^2)t^2\}$, the MGF of $N(m_1 + m_2, s_1^2 + s_2^2)$. Applying this to $y = \theta + \varepsilon$ immediately gives $y \sim N(\mu_0, \tau_0^2 + \sigma^2)$. $\square$

Key Idea

Marginally, y is centered at the prior mean μ_0 but is *more* spread out than either θ or the noise alone: its variance is the *sum* of two independent sources of uncertainty. This additivity of variances for independent sums is the same fact that powers the multivariate normal construction in the next section.

4.4 The Multivariate Normal Approach

The results above can be re-derived – and, more importantly, *unified* – by recognizing that $(\theta, y_1, \dots, y_n)$ is jointly Gaussian. This viewpoint is more work for this simple model but generalizes seamlessly to vector-valued θ , correlated noise, and linear regression with Gaussian priors (Kalman filters, Gaussian process regression), where direct completing-the-square computations become unwieldy.

Definition 4.4 (Multivariate normal). $Z \sim \text{MVN}(m, \Sigma)$ on $\mathbb{R}^k$ has density $p(z) \propto \exp\{-\frac{1}{2}(z - m)^\top \Sigma^{-1}(z - m)\}$ for positive-definite Σ .

Theorem 4.5 (Two standard MVN facts). (a) (Affine transformations.) If $Z \sim \text{MVN}(m, \Sigma)$ and $Y = AZ + c$, then $Y \sim \text{MVN}(Am + c, A\Sigma A^\top)$.

(b) (Conditioning.) Partition $Z = \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} \sim \text{MVN}\left(\begin{pmatrix} m_1 \\ m_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}\right)$. Then $Z_1 | Z_2 = z_2 \sim \text{MVN}(m_1 + \Sigma_{12}\Sigma_{22}^{-1}(z_2 - m_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$.

(Standard results: (a) follows from the MGF of Z ; (b) follows from block-inverting Σ via the Schur complement and completing the square in the joint density's quadratic form. We use both as computational tools.)

Constructing the joint vector. With $\theta \sim N(\mu_0, \tau_0^2)$, $y_i = \theta + \varepsilon_i$, $\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$ independent of θ , stack the *primitive* independent pieces $\phi = (\theta, \varepsilon_1, \dots, \varepsilon_n)^\top$, which is jointly Gaussian with diagonal covariance since its components are mutually independent. The vector $(\theta, y_1, \dots, y_n)^\top = A\phi$ for the linear map A whose first row selects θ and whose $(i+1)$ -th row computes $\theta + \varepsilon_i = y_i$. By Theorem 4.5(a), $(\theta, y_1, \dots, y_n)^\top$ is jointly Gaussian with mean $(\mu_0, \dots, \mu_0)^\top$ and, computing covariances directly (using independence of θ and the ε_i 's, and mutual independence of the ε_i 's):

$$\text{Var}(\theta) = \tau_0^2, \quad \text{Var}(y_i) = \tau_0^2 + \sigma^2, \quad \text{Cov}(\theta, y_i) = \tau_0^2, \quad \text{Cov}(y_i, y_j) = \tau_0^2 \ (i \neq j).$$

Every pair of y_i, y_j shares covariance τ_0^2 – the “common” randomness contributed by the shared θ – exactly matching Theorem 4.3's marginal variance $\tau_0^2 + \sigma^2$ for each individual y_i .

Recovering the posterior. Applying Theorem 4.5(b) with $Z_1 = \theta$, $Z_2 = y$, and simplifying the resulting expression via the Sherman–Morrison identity for the rank-one-perturbed matrix $\Sigma_{22} = \sigma^2 I_n + \tau_0^2 \mathbf{1}_n \mathbf{1}_n^\top$ yields, after algebra,

$$\begin{aligned} E[\theta \mid y] &= \mu_0 + \frac{n\tau_0^2}{\sigma^2 + n\tau_0^2}(\bar{y} - \mu_0) = \frac{\sigma^2\mu_0 + n\tau_0^2\bar{y}}{\sigma^2 + n\tau_0^2} = \frac{\mu_0/\tau_0^2 + n\bar{y}/\sigma^2}{1/\tau_0^2 + n/\sigma^2} = \mu_1, \\ \text{Var}(\theta \mid y) &= \tau_0^2 - \tau_0^4 \cdot \frac{n}{\sigma^2 + n\tau_0^2} = \frac{\tau_0^2\sigma^2}{\sigma^2 + n\tau_0^2} = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}\right)^{-1} = \tau_1^2, \end{aligned}$$

exactly reproducing Theorem 4.2.

Connecting the Threads

The MVN derivation is more algebra for this simple model, but its conceptual payoff is large: *the entire Bayesian update – both the marginal of y and the posterior of $\theta \mid y$ – falls out of one single fact, that (θ, y) is jointly Gaussian.* This is the same structural idea that underlies Gibbs sampling in Chapter 10: once you know how to condition one block of a joint distribution on another, updating becomes mechanical regardless of how many parameters are involved.

4.5 Summary Tables

	Beta–Bernoulli	Normal–Normal (known σ^2)
Model	$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta)$, $\theta \sim \text{Beta}(\alpha, \beta)$	$y_1, \dots, y_n \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$, $\theta \sim N(\mu_0, \tau_0^2)$
Posterior	$\text{Beta}(\alpha + \sum x_i, \beta + n - \sum x_i)$	$N(\mu_1, \tau_1^2)$, precision-weighted as above
Posterior mean	$\frac{\alpha + \sum x_i}{\alpha + \beta + n}$ (weighted avg. of prior mean and MLE $\bar{x}$)	$\mu_1 = \frac{\mu_0/\tau_0^2 + n\bar{y}/\sigma^2}{1/\tau_0^2 + n/\sigma^2}$
Posterior variance	$\rightarrow 0$ as $n \rightarrow \infty$ (rate $1/n$)	$\tau_1^2 = \sigma^2/n \rightarrow 0$ as $n \rightarrow \infty$
Sufficient statistic	$\sum_i x_i$	$\bar{y} = \frac{1}{n} \sum y_i$
Marginal of data	z uniform on $\{0, \dots, n\}$ (flat prior only)	$y \sim N(\mu_0, \tau_0^2 + \sigma^2)$

Recipe

The conjugacy recipe, in general.

1. Write $p(\theta \mid \text{data}) \propto p(\theta) p(\text{data} \mid \theta)$ and drop every multiplicative factor that does not involve θ .
2. Pattern-match the remaining expression in θ to a known density family: powers of θ and $1 - \theta$ signal a Beta kernel; a quadratic exponent in θ signals a Normal kernel (complete the square).
3. The normalizing constant is automatically that of the recognized family – the integral $\int p(\theta) p(\text{data} \mid \theta) d\theta$ never needs to be computed directly.

Part III

Improper Priors, Jeffreys Priors, and Poisson Regression

5 Improper Priors

Every prior used so far – uniform, Beta, Normal – was a genuine probability distribution. This chapter asks whether that is actually necessary, and answers: no, for the prior; yes, for the posterior.

5.1 Does the Prior Need to Integrate to 1?

Key Idea

The prior need not be proper. The posterior must be.

Bayes' rule is used only through the proportionality $\pi(\theta | x) \propto \pi(\theta)f(x | \theta)$, and this remains meaningful even when $\int \pi(\theta) d\theta = \infty$, so long as the right-hand side is integrable in θ – i.e., it can be renormalized into a genuine density. Concretely: $\pi(\theta)$ need only satisfy $\pi(\theta) \geq 0$; but we must have $\int \pi(\theta) f(x | \theta) d\theta < \infty$ so that $\pi(\theta | x)$ is a bona fide, normalizable density.

Definition 5.1 (Improper prior). $\pi(\theta)$ is an *improper prior* if $\pi(\theta) \geq 0$ but $\int \pi(\theta) d\theta = \infty$. It is used only through its shape, $\pi(\theta) \propto \tilde{\pi}(\theta)$, since any multiplicative constant cancels between the numerator and denominator of Bayes' rule.

Theorem 5.2 (Propriety condition). *Suppose $\pi(\theta) \propto \tilde{\pi}(\theta)$ is improper. The posterior from data x is proper if and only if $\int \tilde{\pi}(\theta) f(x | \theta) d\theta < \infty$.*

Proof. The unnormalized posterior kernel is $\tilde{\pi}(\theta)f(x | \theta)$. This renormalizes to a probability density exactly when its total integral over θ is finite and positive: if $\int \tilde{\pi}(\theta)f(x | \theta) d\theta = C < \infty$, then $\pi(\theta | x) = \tilde{\pi}(\theta)f(x | \theta)/C$ integrates to 1. If $C = \infty$, no renormalization is possible. $\square$

Example 5.3 (Flat prior with Normal likelihood – a proper posterior despite an improper prior). Let $X | \theta \sim N(\theta, 1)$ and $\pi(\theta) \propto 1$ on $\mathbb{R}$ (improper, since $\int_{-\infty}^{\infty} 1 d\theta = \infty$). After observing x ,

$$\pi(\theta | x) \propto \exp \left\{ -\frac{(\theta - x)^2}{2} \right\},$$

exactly the kernel of $N(x, 1)$, which is proper. So $\theta | x \sim N(x, 1)$ – this is also the $\tau_0^2 \rightarrow \infty$ limit of Theorem 4.1 with $n = 1$, $\sigma^2 = 1$.

Example 5.4 (When the choice of improper prior matters). Let $X | \theta \sim \text{Uniform}(0, \theta)$, so $f(x | \theta) = \frac{1}{\theta} \mathbf{1}\{0 < x < \theta\}$. With the flat improper prior $\pi(\theta) \propto 1$, $\pi(\theta | x) \propto \frac{1}{\theta} \mathbf{1}\{\theta > x\}$, but $\int_x^{\infty} \theta^{-1} d\theta = \infty$: this posterior is *improper* and unusable. Trying instead $\pi(\theta) \propto 1/\theta$ gives $\pi(\theta | x) \propto \theta^{-2} \mathbf{1}\{\theta > x\}$, and $\int_x^{\infty} \theta^{-2} d\theta = 1/x < \infty$ for $x > 0$: this posterior, $\pi(\theta | x) = x\theta^{-2} \mathbf{1}\{\theta > x\}$ (a Pareto-type density), is proper. Not every “vague” improper prior yields a usable posterior – propriety must be checked case by case.

Recipe

Whenever an improper prior is used, verify $\int \pi(\theta)f(x | \theta) d\theta < \infty$ after seeing the data (and typically for all n at or above some minimal sample size) before trusting the “posterior” as a genuine probability distribution.

6 Jeffreys Priors and Fisher Information

6.1 Why the Flat Prior Is Not Really Non-Informative

A flat prior $\pi(\theta) \propto 1$ looks maximally “uninformative,” but this is an illusion: flatness is not invariant under reparametrization. If $\phi = h(\theta)$ is a smooth, monotone, one-to-one transformation, the change-of-variables (Jacobian) formula gives the induced density

$$p(\phi) = \pi(\theta) \left| \frac{d\theta}{d\phi} \right| = \left| \frac{d\theta}{d\phi} \right|,$$

which is generally *not* flat in ϕ . So “no information about θ ” and “no information about $\phi = h(\theta)$ ” are incompatible statements whenever we default to flatness – an inconsistency for a supposedly ignorance-encoding prior, since the choice of parametrization is entirely arbitrary.

Key Idea

Jeffreys’ principle asks for a rule that produces an *equivalent* prior whether applied directly to θ , or applied to a reparametrization $\phi = h(\theta)$ and then transformed back via the Jacobian. The two paths must agree.

6.2 Fisher Information and the Jeffreys Prior

Definition 6.1 (Fisher information). For $X \mid \theta \sim f(x \mid \theta)$ with log-likelihood $\ell(\theta) = \log f(x \mid \theta)$,

$$I(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2 \ell(\theta)}{\partial \theta^2} \right] = \mathbb{E}_\theta \left[\left(\frac{\partial \ell(\theta)}{\partial \theta} \right)^2 \right].$$

Proposition 6.2 (Equivalence of the two forms). *Under regularity conditions permitting differentiation under the integral sign, $\mathbb{E}_\theta[(\partial \ell / \partial \theta)^2] = -\mathbb{E}_\theta[\partial^2 \ell / \partial \theta^2]$.*

Proof. Since $\int f(x \mid \theta) dx = 1$ for every θ , differentiate once: $\int \partial f / \partial \theta dx = 0$, i.e. (using $\partial f / \partial \theta = f \cdot \partial \log f / \partial \theta$) $\mathbb{E}_\theta[\partial \log f / \partial \theta] = 0$ – the *score* has mean zero. Differentiate this identity once more with respect to θ :

$$\int \frac{\partial}{\partial \theta} \left(f \cdot \frac{\partial \log f}{\partial \theta} \right) dx = 0 \implies \int \left(\frac{\partial f}{\partial \theta} \cdot \frac{\partial \log f}{\partial \theta} + f \cdot \frac{\partial^2 \log f}{\partial \theta^2} \right) dx = 0.$$

The first term equals $f (\partial \log f / \partial \theta)^2$, so $\mathbb{E}_\theta[(\partial \log f / \partial \theta)^2] + \mathbb{E}_\theta[\partial^2 \log f / \partial \theta^2] = 0$, which is the claimed identity. $\square$

Definition 6.3 (Jeffreys prior). $\pi_J(\theta) \propto \sqrt{I(\theta)}$.

Theorem 6.4 (Invariance of the Jeffreys prior). *Let $\phi = h(\theta)$ be smooth, monotone, and invertible. If $\pi_J(\theta) \propto \sqrt{I(\theta)}$ and $p_J(\phi) \propto \sqrt{I(\phi)}$ (computed directly in the ϕ -parametrization), then $p_J(\phi) = \pi_J(\theta) \left| \frac{d\theta}{d\phi} \right|$: the two priors agree under the ordinary Jacobian transformation.*

Proof. Write $\theta = h^{-1}(\phi)$. By the chain rule, for $\tilde{\ell}(\phi) := \log f(x \mid h^{-1}(\phi))$, $\partial \tilde{\ell} / \partial \phi = (\partial \ell / \partial \theta) \cdot (d\theta / d\phi)$. Squaring and taking expectations (using the information-equality form of Proposition 6.2),

$$I(\phi) = \mathbb{E} \left[\left(\frac{\partial \tilde{\ell}}{\partial \phi} \right)^2 \right] = \mathbb{E} \left[\left(\frac{\partial \ell}{\partial \theta} \right)^2 \right] \left(\frac{d\theta}{d\phi} \right)^2 = I(\theta) \left(\frac{d\theta}{d\phi} \right)^2.$$

Taking square roots, $\sqrt{I(\phi)} = \sqrt{I(\theta)} |d\theta / d\phi|$, i.e. $p_J(\phi) \propto \pi_J(\theta) |d\theta / d\phi|$, exactly the Jacobian transformation rule. $\square$

Key Idea

This invariance is exactly why Jeffreys priors are the standard “objective” default: they form the unique simple construction compatible with parametrization-independent ignorance. Section 6.5 gives the deeper reason this works – Fisher information is the local curvature of the Kullback–Leibler divergence, a genuinely coordinate-free notion of distance between nearby distributions.

6.3 Worked Derivation: Bernoulli/Binomial

Let $X \sim \text{Binomial}(n, p)$. The log-likelihood (dropping the binomial coefficient) is $\ell(p) = X \log p + (n - X) \log(1 - p)$, so $\ell'(p) = X/p - (n - X)/(1 - p)$ and $\ell''(p) = -X/p^2 - (n - X)/(1 - p)^2$. Using $E[X] = np$,

$$I(p) = -E[\ell''(p)] = \frac{np}{p^2} + \frac{n(1-p)}{(1-p)^2} = \frac{n}{p} + \frac{n}{1-p} = \frac{n}{p(1-p)}.$$

Hence $\pi_J(p) \propto \sqrt{1/[p(1-p)]} = p^{-1/2}(1-p)^{-1/2}$ – exactly the kernel of a $\text{Beta}(\frac{1}{2}, \frac{1}{2})$ distribution:

$$p \sim \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right),$$

which is *proper*, since $\int_0^1 p^{-1/2}(1-p)^{-1/2} dp = B(\frac{1}{2}, \frac{1}{2}) = \pi < \infty$.

Connecting the Threads

Compare this to Chapter 3: the Jeffreys prior for a binomial proportion is *itself* a member of the Beta–Bernoulli conjugate family, with $\alpha = \beta = \frac{1}{2}$. This is not a coincidence special to the binomial – Section 6.4 shows the analogous phenomenon for the Poisson/Gamma pair.

6.4 Worked Derivation: Poisson, via the Natural Parameter

The Poisson pmf is $f(x | \lambda) = e^{-\lambda} \lambda^x / x!$, so $\ell(\lambda) = -\lambda + x \log \lambda - \log x!$, $\ell'(\lambda) = -1 + x/\lambda$, $\ell''(\lambda) = -x/\lambda^2$. Since $E[X] = \lambda$,

$$I(\lambda) = -E[\ell''(\lambda)] = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda} \quad \implies \quad \pi_J(\lambda) \propto \lambda^{-1/2}.$$

Recipe

The exponential-family / natural-parameter route to a default prior.

1. Write the likelihood in canonical form $f(x | \eta) \propto \exp\{\eta T(x) - A(\eta)\}$, identifying the natural parameter η and natural statistic $T(x)$.
2. Construct a prior (often flat, or Jeffreys) over η .
3. Use the Jacobian $|d\eta/d\theta|$ to convert this prior back to the original parametrization θ .

For the Poisson, $f(x | \lambda) = \exp\{x \log \lambda - \lambda - \log x!\}$, so $T(x) = x$ and the natural parameter is $\eta = \log \lambda$, i.e. $\lambda = e^\eta$. By Theorem 6.4, transforming $\pi_J(\lambda) \propto \lambda^{-1/2}$ into the η -scale using $|d\lambda/d\eta| = e^\eta = \lambda$ gives $p(\eta) = \lambda^{-1/2} \cdot \lambda = \lambda^{1/2} = e^{\eta/2}$. Checking directly: since $\ell(\eta) = \eta x - e^\eta - \log x!$, $\ell''(\eta) = -e^\eta$, so $I(\eta) = E[e^\eta] = e^\eta$ (a constant given η) and $\sqrt{I(\eta)} = e^{\eta/2}$ – exactly matching $p(\eta)$, a direct numerical confirmation of Theorem 6.4 for the Poisson model.

Since $\int_0^\infty \lambda^{-1/2} d\lambda = \infty$, this Jeffreys prior is improper (as Chapter 5 would lead us to expect for an unbounded parameter space). Its posterior, however, is proper: $\pi(\lambda | x_{1:n}) \propto \lambda^{-1/2} \cdot e^{-n\lambda} \lambda^{\sum x_i} =$

$\lambda \sum x_i - 1/2 e^{-n\lambda}$, exactly a

$$\lambda \mid x_{1:n} \sim \text{Gamma}\left(\sum_i x_i + \frac{1}{2}, n\right)$$

kernel – a clean, proper Gamma posterior, echoing the same “Jeffreys $\Rightarrow$ conjugate-family-adjacent” pattern seen with $\text{Beta}(\frac{1}{2}, \frac{1}{2})$ above.

6.5 KL Divergence and Its Link to Fisher Information

Theorem 6.5 (Local quadratic expansion of KL divergence). *As $\Delta \rightarrow 0$,*

$$D_{\text{KL}}(f(\cdot \mid \theta) \parallel f(\cdot \mid \theta + \Delta)) = \frac{1}{2} I(\theta) \Delta^2 + o(\Delta^2).$$

Proof. Write $\ell(\theta) = \log f(x \mid \theta)$; then $D_{\text{KL}} = \mathbb{E}_\theta[\ell(\theta) - \ell(\theta + \Delta)]$. Taylor-expand $\ell(\theta + \Delta)$ about θ to second order:

$$\ell(\theta + \Delta) = \ell(\theta) + \ell'(\theta)\Delta + \frac{1}{2}\ell''(\theta)\Delta^2 + o(\Delta^2).$$

Taking $-\mathbb{E}_\theta[\cdot]$ of $\ell(\theta + \Delta) - \ell(\theta)$, and using $\mathbb{E}_\theta[\ell'(\theta)] = 0$ (the score has mean zero, Proposition 6.2) and $-\mathbb{E}_\theta[\ell''(\theta)] = I(\theta)$,

$$D_{\text{KL}} = -\left(0 \cdot \Delta + \frac{1}{2}(-I(\theta))\Delta^2\right) + o(\Delta^2) = \frac{1}{2}I(\theta)\Delta^2 + o(\Delta^2). \quad \square$$

Key Idea

$I(\theta)$ measures how quickly the model’s distribution moves away, in KL sense, as θ is perturbed – i.e., how identifiable θ is locally. The Jeffreys prior $\pi_J(\theta) \propto \sqrt{I(\theta)}$ can therefore be read as placing prior mass *uniformly with respect to a natural distance on distributions*, $ds^2 = I(\theta) d\theta^2$ (the Fisher–Rao information metric), rather than uniformly on the arbitrary coordinate θ – which is exactly the non-invariance problem that motivated Jeffreys’ construction in the first place. Chapter 8 develops this metric viewpoint further and proves the Cramér–Rao lower bound, which gives Fisher information yet another interpretation: as a universal speed limit on how precisely θ can be estimated.

Sanity check (Binomial). $I(p) = n/[p(1-p)]$ is large near $p = 0, 1$ and smallest at $p = \frac{1}{2}$ – a coin near 0 or 1 gives sharp, easily identified information from few flips, while $p = \frac{1}{2}$ is hardest to pin down per flip, matching the U-shape of $1/[p(1-p)]$.

7 Poisson Regression with Covariates

We now move from a single Poisson rate λ to a regression setting: for units $i = 1, \dots, n$ with covariates x_i and counts y_i ,

$$Y_i \mid x_i, \theta \stackrel{\text{ind}}{\sim} \text{Poisson}(\lambda(x_i, \theta)), \quad \log \lambda(x_i, \theta) = x_i^\top \theta \iff \lambda(x_i, \theta) = \exp\{x_i^\top \theta\}.$$

The log link guarantees $\lambda > 0$ automatically for any real-valued θ , and is called *canonical* because $\log \lambda$ is precisely the natural parameter of the Poisson exponential family identified in Section 6.4.

7.1 Likelihood and the Bayesian Workflow

$$L(\theta) \propto \exp \left\{ \sum_i y_i x_i^\top \theta - \sum_i e^{x_i^\top \theta} \right\}.$$

Recipe

The complete Bayesian workflow, applicable to any model (Poisson regression included):

1. Write down the likelihood $L(\theta)$.
2. Specify a prior $\pi(\theta)$ – conjugate if available, otherwise Jeffreys/reference.
3. Compute the posterior $\pi(\theta \mid D) \propto \pi(\theta)L(\theta)$.
4. Compute the predictive distribution for a future observation: $p(y^* \mid x^*, D) = \int f(y^* \mid x^*, \theta) \pi(\theta \mid D) d\theta$.

Key Idea

Unlike the single-parameter Poisson model (Gamma-conjugate) or the Normal-mean model (Normal-conjugate), Poisson regression’s log link makes $\lambda_i = e^{x_i^\top \theta}$ a *nonlinear* function of θ , so no finite-dimensional conjugate family exists for θ jointly (except in the degenerate intercept-only case, which reduces to the ordinary Poisson–Gamma model of Section 6.4). In practice one uses a flat improper prior $\pi(\theta) \propto 1$, a Jeffreys/reference prior built from the Fisher information matrix, or a proper but weakly informative prior $\theta \sim N(0, cI)$ for large c (an MCMC-friendly stand-in for “no information”), and explores the posterior numerically.

7.2 The Fisher Information Matrix

For a single observation, $\ell_i(\theta) = y_i x_i^\top \theta - e^{x_i^\top \theta} - \log y_i!$. The score is $\nabla_\theta \ell_i = (y_i - e^{x_i^\top \theta})x_i = (y_i - \lambda_i)x_i$, and the Hessian is $\nabla_\theta^2 \ell_i = -e^{x_i^\top \theta} x_i x_i^\top = -\lambda_i x_i x_i^\top$. Since $E[Y_i] = \lambda_i$ does not affect the Hessian (no y_i remains in it), taking $-E[\cdot]$ and summing over independent observations,

$$I(\theta) = -\sum_{i=1}^n E[\nabla_\theta^2 \ell_i] = \sum_{i=1}^n \lambda_i x_i x_i^\top = X^\top W X, \quad W = \text{diag}(\lambda_1, \dots, \lambda_n),$$

with X the $n \times p$ design matrix with rows $x_i^\top$. This gives the Jeffreys/reference prior $\pi(\theta) \propto \sqrt{\det I(\theta)} = \sqrt{\det(X^\top W X)}$, the multivariate generalization of $\pi_J(\theta) \propto \sqrt{I(\theta)}$ from Chapter 6.

7.3 Summary Table

Model	Likelihood	Fisher info $I(\theta)$	Jeffreys prior
Normal mean, σ^2 known	$N(\theta, \sigma^2)$	$1/\sigma^2$ (const.)	$\pi(\theta) \propto 1$ (flat)
Binomial	$\text{Binomial}(n, p)$	$n/[p(1 - p)]$	$\text{Beta}(\frac{1}{2}, \frac{1}{2})$
Poisson	$\text{Poisson}(\lambda)$	$1/\lambda$	$\lambda^{-1/2}$ (improper; proper posterior)
Poisson regression	$\prod \text{Poisson}(e^{x_i^\top \theta})$	$X^\top W X$	$\propto \sqrt{\det(X^\top W X)}$

Connecting the Threads

This table is deliberately incomplete – it is the seed of a much larger table. Chapter 9 rederives the Bernoulli and Poisson rows from scratch (as an independent check), adds two Normal-variance rows, and folds everything into a single master reference in Section 9.1.

Part IV

Fisher Information and Multi-Parameter Models

8 The Cramér–Rao Bound

Part III introduced Fisher information as the ingredient of the Jeffreys prior. This chapter proves its other classical role: a universal lower bound on the variance of any unbiased estimator, and derives the total information for a sample of size n .

Definition 8.1. For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(x | \theta)$, the total information is $I_n(\theta) = nI(\theta)$, since log-likelihoods add across independent observations and expectations are linear.

Theorem 8.2 (Cramér–Rao Lower Bound). *Let $T(X)$ be any unbiased estimator of θ based on $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(x | \theta)$. Then*

$$\text{Var}_\theta(T(X)) \geq \frac{1}{I_n(\theta)} = \frac{1}{nI(\theta)}.$$

Proof. Let $S(\theta) = \sum_{i=1}^n \partial \log f(X_i | \theta) / \partial \theta$ be the total score, so $\mathbb{E}[S] = 0$ and $\text{Var}(S) = I_n(\theta)$ (a sum of n i.i.d. mean-zero terms, each of variance $I(\theta)$). Since T is unbiased, $\mathbb{E}_\theta[T(X)] = \theta$ for all θ ; differentiating with respect to θ ,

$$\frac{d}{d\theta} \int T(x) f(x | \theta) dx = 1 \implies \int T(x) f(x | \theta) S(\theta) dx = 1 \implies \text{Cov}(T, S) = 1.$$

By Cauchy–Schwarz, $\text{Cov}(T, S)^2 \leq \text{Var}(T) \text{Var}(S)$, i.e. $1 \leq \text{Var}(T) \cdot I_n(\theta)$, so $\text{Var}(T) \geq 1/I_n(\theta)$. $\square$

An estimator attaining the bound with equality is called *efficient*.

Example 8.3 (Efficiency of $\bar{X}$ for a binomial proportion). For $\text{Binomial}(n, p)$, the natural unbiased estimator is $T(X) = X/n = \bar{X}$, with $\text{Var}(T) = p(1-p)/n$. The CRLB from Theorem 8.2 is $1/I(p) = p(1-p)/n$ (using $I(p) = n/[p(1-p)]$ from Chapter 6, so the *per-observation* information is $p(1-p)^{-1}$... more precisely $I_1(p) = 1/[p(1-p)]$ per Bernoulli trial, giving $1/I_n(p) = p(1-p)/n$). These coincide exactly: $\bar{X}$ is an efficient estimator of p , attaining the Cramér–Rao bound with equality.

8.1 Fisher Information as Curvature, Revisited

Section 6.5 already proved the local quadratic expansion $D_{\text{KL}}(f(\cdot | \theta) \| f(\cdot | \theta + \Delta)) = \frac{1}{2} I(\theta) \Delta^2 + o(\Delta^2)$. Combined with the Cramér–Rao bound just proved, we now have *two* independent operational meanings for the same quantity $I(\theta)$:

Key Idea

Fisher information is simultaneously (i) the local curvature of the KL divergence between nearby members of a parametric family – how distinguishable close values of θ are – and (ii) the reciprocal of the best possible variance of an unbiased estimator – how precisely θ can be pinned down from data. These are two views of the same underlying geometric fact, and together they justify treating $\sqrt{I(\theta)}$ as a natural “information-weighted” prior density: it is exactly proportional to local estimability.

9 Jeffreys Priors: A Complete Reference

This chapter re-derives, side by side and in a uniform notation, every Jeffreys prior needed in this book, so that Part V can simply cite the results. Re-deriving the Bernoulli and Poisson cases (already done once in Chapter 6) also serves as an independent consistency check on Theorem 6.4.

Exercise 9.1 (Poisson(λ)). Re-derive $\pi_J(\lambda)$, and verify invariance by transforming to $\phi = \sqrt{\lambda}$.

Solution. As in Section 6.4, $I(\lambda) = 1/\lambda$, so $\pi_J(\lambda) \propto \lambda^{-1/2}$. For the invariance check, let $\phi = \sqrt{\lambda}$, so $\lambda = \phi^2$ and $d\lambda/d\phi = 2\phi$. Transforming, $\pi_J(\phi) \propto \lambda^{-1/2} \cdot 2\phi = \phi^{-1} \cdot 2\phi = 2 \propto 1$ – flat in $\phi = \sqrt{\lambda}$. This is a clean illustration of invariance: the Jeffreys prior on λ is precisely the one that makes $\sqrt{\lambda}$ -space flat. $\square$

Exercise 9.2 (Binomial(n, p)). Re-derive $\pi_J(p)$ and verify efficiency of $\bar{X} = X/n$.

Solution. As in Chapter 6, $I(p) = n/[p(1-p)]$, so $\pi_J(p) \propto p^{-1/2}(1-p)^{-1/2}$, the Beta($\frac{1}{2}, \frac{1}{2}$) kernel – also known as the *arcsine distribution*, the standard reference prior for a binomial proportion. Efficiency of $\bar{X}$ was verified in the Example above. $\square$

Exercise 9.3 (Normal, mean unknown, variance known). $X \sim N(\mu, \sigma^2)$, σ^2 known. Derive $\pi_J(\mu)$.

Solution. $\log f(x | \mu) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}$, so $\partial \log f / \partial \mu = (x - \mu)/\sigma^2$ and $\partial^2 \log f / \partial \mu^2 = -1/\sigma^2$ – already a constant, with no expectation needed: $I(\mu) = 1/\sigma^2$, not depending on μ . Hence $\pi_J(\mu) \propto \sqrt{1/\sigma^2} \propto 1$, the flat (improper) prior. This is the one situation encountered so far where the flat prior and the Jeffreys prior coincide exactly, precisely because $I(\mu)$ is constant in μ – a hallmark of *location-family* structure, formalized independently in Section 13.1. $\square$

Exercise 9.4 (Normal, mean known, variance unknown). $X \sim N(\mu, \sigma^2)$, μ known, $\theta := \sigma^2$ unknown. Derive $\pi_J(\theta)$.

Solution. $\log f(x | \theta) = -\frac{1}{2} \log(2\pi\theta) - \frac{(x-\mu)^2}{2\theta}$, so $\partial \log f / \partial \theta = -\frac{1}{2\theta} + \frac{(x-\mu)^2}{2\theta^2}$ and $\partial^2 \log f / \partial \theta^2 = \frac{1}{2\theta^2} - \frac{(x-\mu)^2}{\theta^3}$. Using $E[(X - \mu)^2] = \theta$,

$$I(\theta) = -\left(\frac{1}{2\theta^2} - \frac{\theta}{\theta^3}\right) = -\left(\frac{1}{2\theta^2} - \frac{1}{\theta^2}\right) = \frac{1}{2\theta^2}.$$

Hence $\pi_J(\sigma^2) \propto \sqrt{1/(2\sigma^4)} \propto 1/\sigma^2$. Equivalently, in terms of σ itself (via Theorem 6.4, with $\theta = \sigma^2$, $d\theta/d\sigma = 2\sigma$): $\pi_J(\sigma) \propto \sigma^{-2} \cdot 2\sigma \propto 1/\sigma$. Both are the standard “scale prior” $\propto 1/(\text{scale})$ that recurs throughout objective Bayesian analysis – it is precisely the unique prior making $\log \sigma^2$ (or $\log \sigma$) flat, the natural invariance for a positive scale parameter. $\square$

9.1 Summary Table of Common Jeffreys Priors

Model	Fisher info $I(\theta)$	Jeffreys prior $\pi_J(\theta)$	Proper?
Poisson(λ)	$1/\lambda$	$\lambda^{-1/2}$	No (posterior is)
Binomial(n, p)	$n/[p(1-p)]$	Beta($\frac{1}{2}, \frac{1}{2}$)	Yes
$N(\mu, \sigma^2)$, σ^2 known	$1/\sigma^2$ (const. in μ)	$\propto 1$ (flat)	No
$N(\mu, \sigma^2)$, μ known	$1/(2\sigma^4)$	$\propto 1/\sigma^2$	No
Poisson regression, $\theta \in \mathbb{R}^p$	$X^\top W X$	$\propto \sqrt{\det(X^\top W X)}$	No

Connecting the Threads

Look at the pattern across every row: a *location* parameter (μ with σ^2 known) gets a flat prior; a *scale* parameter (σ^2 with μ known) gets a $1/(\text{scale})$ prior. Chapter 13 proves this pattern is not a coincidence of the Normal family specifically – it follows from an entirely different argument (invariance under a pivotal quantity, with no Fisher information anywhere in sight) that arrives at exactly these same two priors for *any* location or scale family.

9.2 Non-Informative Priors: A Taxonomy

Type	Description
Flat	$\pi(\theta) \propto 1$. Simple, but not invariant under reparametrization (Theorem 6.4 shows the failure mode precisely), and can be surprisingly informative in disguise (flat in σ^2 is very different from flat in $\log \sigma^2$).
Jeffreys	$\pi_J(\theta) \propto \sqrt{I(\theta)}$. Automatically invariant; the standard default recipe. Often improper but usually yields a proper posterior (Chapter 5).
Diffuse/vague	A convenient proper prior (e.g. $N(\mu_0, \sigma_0^2)$) with a hyperparameter sent to its non-informative limit (e.g. $\sigma_0^2 \rightarrow \infty$), recovering the flat-prior case exactly, as in Corollary 12.1. Popular because it stays inside a familiar conjugate family right up to the limit, and provides a useful sanity check on Jeffreys-prior calculations.

None of these is “no information” in an absolute sense – improper priors are not honest probability distributions, and every choice imposes some structure. The practical justification is asymptotic: by the Bernstein–von Mises theorem (discussed further in Section 14.1), the choice among reasonable non-informative priors has a vanishing effect on the posterior as $n \rightarrow \infty$, while Jeffreys’ invariance makes it the most defensible default when no other information is available.

10 Multi-Parameter Bayesian Models

Every model so far has had a single unknown parameter. We now consider $\theta = (\theta_1, \theta_2)$, e.g. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(x | \theta_1, \theta_2)$. Bayes' rule extends immediately:

$$\pi(\theta_1, \theta_2 | D) \propto \pi(\theta_1, \theta_2) f(D | \theta_1, \theta_2), \quad D = (X_1, \dots, X_n).$$

10.1 Nuisance Parameters and Marginalization

Definition 10.1 (Nuisance parameter). If only θ_1 is of direct interest and θ_2 must be modeled only because it affects the likelihood, θ_2 is a *nuisance parameter*. The Bayesian solution is to integrate it out: $\pi(\theta_1 | D) = \int \pi(\theta_1, \theta_2 | D) d\theta_2$.

Key Idea

This is conceptually the cleanest feature of the Bayesian treatment of nuisance parameters, compared to the more delicate frequentist devices (profile likelihood, conditioning, ancillarity): once the joint posterior is known, marginalizing is *just integration*, no extra machinery required.

Proposition 10.2. Writing $\pi(\theta_1, \theta_2 | D) = \pi(\theta_1 | \theta_2, D) \pi(\theta_2 | D)$ (the ordinary conditional-density factorization),

$$\pi(\theta_1 | D) = \int \pi(\theta_1 | \theta_2, D) \pi(\theta_2 | D) d\theta_2 :$$

the marginal posterior of θ_1 is a $\pi(\theta_2 | D)$ -weighted mixture of the conditional posteriors $\pi(\theta_1 | \theta_2, D)$.

10.2 Monte Carlo Marginalization

Analytically computing $\int \pi(\theta_1, \theta_2 | D) d\theta_2$ is often intractable. The simulation-based alternative:

Proposition 10.3. If $(\theta_1^{(b)}, \theta_2^{(b)}) \stackrel{\text{iid}}{\sim} \pi(\theta_1, \theta_2 | D)$ for $b = 1, \dots, B$, then $\theta_1^{(1)}, \dots, \theta_1^{(B)}$ – obtained simply by discarding the θ_2 coordinates – is itself a sample from the marginal posterior $\pi(\theta_1 | D)$.

Proof. This is the defining property of a marginal distribution: if (Θ_1, Θ_2) has joint density $\pi(\theta_1, \theta_2)$, then by definition the marginal density of Θ_1 is $\int \pi(\theta_1, \theta_2) d\theta_2$, and the projection of a joint random draw onto its first coordinate has, as its own law, exactly that marginal. Recording the $\theta_1^{(b)}$'s and forming their empirical distribution therefore consistently estimates $\pi(\theta_1 | D)$ (by the Glivenko–Cantelli theorem), without ever writing the marginalization integral explicitly. $\square$

Key Idea

This converts a possibly-hard analytic integration problem into an often-easy simulation and bookkeeping problem – *provided* we have a way to draw from the joint posterior in the first place. Gibbs sampling, described next, supplies exactly that.

10.3 Worked Example: Both μ and σ^2 Unknown

Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ with both μ and σ^2 unknown, and we want $\pi(\mu | D)$.

- If σ^2 were known, ordinary single-parameter conjugate theory (Theorem 4.2 or Exercise 9.3) gives $\pi(\mu \mid \sigma^2, D)$ directly.
- We also separately need $\pi(\sigma^2 \mid D)$ – the marginal posterior of the variance alone, which turns out to be a *scaled inverse- χ^2* distribution (developed in Section 11 and used in Chapter 12).
- What we actually want, $\pi(\mu \mid D) = \int \pi(\mu \mid \sigma^2, D) \pi(\sigma^2 \mid D) d\sigma^2$, is exactly the nuisance-parameter integral above, with $\theta_1 = \mu, \theta_2 = \sigma^2$.

Recipe

Two-step Monte Carlo sampling scheme.

Step 1. Draw $\sigma^{2(b)}$ from the marginal posterior $\sigma^2 \mid D$.

Step 2. Draw $\mu^{(b)} \mid \sigma^{2(b)}, D$ using the known conditional posterior formula for μ given a fixed σ^2 .

Repeating for $b = 1, \dots, B$ produces joint draws $(\mu^{(b)}, \sigma^{2(b)})$ from the joint posterior, and by Proposition 10.3 the $\mu^{(b)}$'s alone form a sample from the desired marginal $\pi(\mu \mid D)$ – without ever evaluating the mixing integral in closed form.

This two-step scheme is the simplest possible instance of *Gibbs sampling*, formalized next; Chapter 12 carries it out with every distribution made fully explicit.

10.4 Gibbs Sampling

Definition 10.4 (Gibbs sampler). Given a joint posterior $\pi(\theta_1, \theta_2 \mid D)$ that is hard to sample from directly, but whose two full conditionals $\pi(\theta_1 \mid \theta_2, D)$ and $\pi(\theta_2 \mid \theta_1, D)$ are each easy to sample from, the Gibbs sampler generates a Markov chain $(\theta_1^{(t)}, \theta_2^{(t)})_{t \geq 1}$ by alternating

$$\theta_2^{(t)} \sim \pi(\theta_2 \mid \theta_1^{(t-1)}, D), \quad \theta_1^{(t)} \sim \pi(\theta_1 \mid \theta_2^{(t)}, D),$$

and repeating.

Key Idea

Why it works (sketch). Each conditional draw uses the most recently sampled value of the other coordinate. Under mild regularity conditions, the two full conditionals uniquely determine the joint distribution they came from (the Hammersley–Clifford/Besag factorization theorem), and iterating the two conditional draws is a special case of the Metropolis–Hastings algorithm with acceptance probability identically 1 – hence a valid MCMC chain converging in distribution to $\pi(\theta_1, \theta_2 \mid D)$ as $t \rightarrow \infty$. After burn-in, the retained draws behave like (correlated) samples from the joint posterior, and Proposition 10.3's “ignore the other coordinate” trick applies exactly as before to extract any desired marginal.

Nothing about Gibbs sampling is special to the Normal model – it applies whenever a joint posterior over several parameters (or blocks of parameters) has tractable full conditionals, even when the joint density itself has no closed form. It is one of the two workhorse MCMC algorithms (with Metropolis–Hastings) for Bayesian computation.

11 The Scaled Inverse- χ^2 Distribution

This is the distribution that appears as the marginal posterior of σ^2 in the Normal model of Section 10.3, and is the last piece of machinery needed before Part V can complete that derivation in full.

Definition 11.1. $X \sim \text{Scaled-Inv-}\chi^2(\nu, \tau^2)$ if its density is

$$f(x; \nu, \tau^2) = \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} \tau^\nu x^{-(\nu/2+1)} \exp\left\{-\frac{\nu\tau^2}{2x}\right\}, \quad x > 0.$$

Proposition 11.2 (Three standard relationships). (a) If $X \sim \text{Scaled-Inv-}\chi^2(\nu, \tau^2)$ then $\nu\tau^2/X \sim \chi_\nu^2$.

(b) $X \sim \text{Inv-Gamma}(\frac{\nu}{2}, \frac{\nu\tau^2}{2})$.

(c) For $k > 0$, $kX \sim \text{Scaled-Inv-}\chi^2(\nu, k\tau^2)$.

Proof. (a) Let $Y = \nu\tau^2/X$, so $X = \nu\tau^2/Y$ and $|dx/dy| = \nu\tau^2/y^2$. By the change-of-variables formula,

$$f_Y(y) = \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} \tau^\nu \left(\frac{\nu\tau^2}{y}\right)^{-(\nu/2+1)} \exp\left\{-\frac{\nu\tau^2}{2} \cdot \frac{y}{\nu\tau^2}\right\} \cdot \frac{\nu\tau^2}{y^2}.$$

Collecting powers of y : from $(\nu\tau^2/y)^{-(\nu/2+1)}$ we get $y^{\nu/2+1}(\nu\tau^2)^{-(\nu/2+1)}$; multiplying by the extra $\nu\tau^2/y^2$ factor gives $y^{\nu/2-1}(\nu\tau^2)^{-\nu/2}$. So

$$f_Y(y) = \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} \tau^\nu \cdot y^{\nu/2-1} (\nu\tau^2)^{-\nu/2} e^{-y/2} = \frac{1}{2^{\nu/2} \Gamma(\nu/2)} y^{\nu/2-1} e^{-y/2},$$

using $\tau^\nu \cdot \tau^{-\nu} = 1$ and $(\nu/2)^{\nu/2}/\nu^{\nu/2} = (1/2)^{\nu/2}$. This is exactly the χ_ν^2 density.

(b) Recall $\text{Inv-Gamma}(\alpha, \beta)$ has density $f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} e^{-\beta/x}$. Rewriting the scaled-inv- χ^2 constant as $(\nu/2)^{\nu/2} \tau^\nu = (\nu\tau^2/2)^{\nu/2}$ shows $f(x; \nu, \tau^2)$ matches $\text{Inv-Gamma}(\alpha, \beta)$ exactly with $\alpha = \nu/2$, $\beta = \nu\tau^2/2$.

(c) Let $Z = kX$, so $X = Z/k$, $|dx/dz| = 1/k$. Then

$$f_Z(z) = \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} \tau^\nu \left(\frac{z}{k}\right)^{-(\nu/2+1)} \exp\left\{-\frac{\nu\tau^2 k}{2z}\right\} \cdot \frac{1}{k}.$$

Collecting powers of k : $(z/k)^{-(\nu/2+1)} = z^{-(\nu/2+1)} k^{\nu/2+1}$, and multiplying by the extra $1/k$ gives $k^{\nu/2}$. So

$$f_Z(z) = \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} (k\tau^2)^{\nu/2} z^{-(\nu/2+1)} \exp\left\{-\frac{\nu(k\tau^2)}{2z}\right\},$$

using $\tau^\nu k^{\nu/2} = (k\tau^2)^{\nu/2}$ – exactly the $\text{Scaled-Inv-}\chi^2(\nu, k\tau^2)$ density. □

Key Idea

In the conjugate Bayesian analysis of $N(\mu, \sigma^2)$ with the standard non-informative prior $\pi(\mu, \sigma^2) \propto 1/\sigma^2$, the marginal posterior of σ^2 works out to be exactly a scaled inverse- χ^2 distribution, with ν related to the sample size and τ^2 related to the sample variance. Chapter 12 derives this explicitly and shows it is exactly what Step 1 of the two-step sampler in Section 10.3 draws from.

11.1 Consolidated Summary of Part IV

- Fisher information $I(\theta) = E[(\partial \log f / \partial \theta)^2] = -E[\partial^2 \log f / \partial \theta^2]$ measures the curvature of expected log-likelihood / local KL divergence, and lower-bounds the variance of unbiased estimators (Cramér–Rao).
- The Jeffreys prior $\pi_J(\theta) \propto \sqrt{I(\theta)}$ is the unique simple recipe – among flat, Jeffreys, and diffuse priors – that is invariant to reparametrization.
- Worked Jeffreys priors: $\text{Poisson}(\lambda) \Rightarrow \lambda^{-1/2}$; $\text{Binomial}(n, p) \Rightarrow \text{Beta}(\frac{1}{2}, \frac{1}{2})$; $N(\mu, \sigma^2)$ (σ^2 known) $\Rightarrow$ flat; $N(\mu, \sigma^2)$ (μ known) $\Rightarrow 1/\sigma^2$.
- In multi-parameter models, nuisance parameters are handled by marginalizing the joint posterior; when the integral is intractable, Monte Carlo marginalization and Gibbs sampling provide computational routes to the same answer.
- The scaled inverse- χ^2 distribution – related to χ_ν^2 , to the inverse-gamma, and closed under positive scaling – is the standard conjugate marginal posterior for a Normal variance.

Part V

The Full Normal Model, Reference Priors, MLE/MAP, and Credible Intervals

12 The Normal Model with Both μ and σ^2 Unknown

We now carry out, completely, the calculation previewed in Section 10.3: $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ with both parameters unknown.

12.1 The Independent Jeffreys Prior

Take μ and σ^2 to be a priori independent, each given its own marginal Jeffreys prior from Exercises 9.3–9.4: $\pi(\mu) \propto 1$ and $\pi(\sigma^2) \propto 1/\sigma^2$. Their product is

$$\pi(\mu, \sigma^2) = \pi(\mu) \pi(\sigma^2) \propto \frac{1}{\sigma^2}.$$

Key Idea

This *independence* construction – building $\pi(\theta_1, \theta_2) \propto \pi(\theta_1 | \cdot) \pi(\theta_2 | \cdot)$ by treating each parameter as if it alone were unknown, then multiplying – is the standard textbook default, but it is *not* in general identical to the true joint Jeffreys prior $\pi(\mu, \sigma^2) \propto \sqrt{\det I(\mu, \sigma^2)}$ for the two-parameter Normal family (which instead gives $\propto 1/\sigma^3$). The independence prior is used here because it also matches the scale-invariance argument of Chapter 13, applied separately to each parameter.

The joint posterior is

$$\pi(\mu, \sigma^2 | y) \propto \frac{1}{\sigma^2} \cdot (\sigma^2)^{-n/2} \exp \left\{ -\frac{\sum_i (y_i - \mu)^2}{2\sigma^2} \right\} = \sigma^{-n-2} \exp \left\{ -\frac{\sum_i (y_i - \mu)^2}{2\sigma^2} \right\}.$$

12.2 Step 1: The Conditional Posterior of μ Given σ^2

We use the factorization $\pi(\mu, \sigma^2 | y) = \pi(\mu | \sigma^2, y) \pi(\sigma^2 | y)$, and the key simplification: conditional on σ^2 , it behaves like a known constant, so $\pi(\mu | \sigma^2, y)$ follows from ordinary single-parameter conjugate theory (Theorem 4.2).

Corollary 12.1 (Flat prior as a limit of Normal–Normal conjugacy). *In Theorem 4.2, let $\tau_0^2 \rightarrow \infty$ (an infinitely diffuse prior on μ) with μ_0 fixed. Then $\mu_1 \rightarrow \bar{y}$ and $\tau_1^2 \rightarrow \sigma^2/n$.*

Proof. As $\tau_0^2 \rightarrow \infty$, $1/\tau_0^2 \rightarrow 0$, so $A := 1/\tau_0^2 + n/\sigma^2 \rightarrow n/\sigma^2$ and $B := \mu_0/\tau_0^2 + n\bar{y}/\sigma^2 \rightarrow n\bar{y}/\sigma^2$ (the μ_0/τ_0^2 term vanishes regardless of μ_0). Hence $\tau_1^2 = 1/A \rightarrow \sigma^2/n$ and $\mu_1 = B/A \rightarrow \bar{y}$. $\square$

This gives, directly, the conditional posterior under the flat/Jeffreys prior on μ :

$$\mu | \sigma^2, y \sim N\left(\bar{y}, \frac{\sigma^2}{n}\right). \quad (\text{R1})$$

12.3 Step 2: The Marginal Posterior of σ^2

To obtain $\pi(\sigma^2 | y)$ alone, integrate the nuisance parameter μ out of the joint posterior:

$$\pi(\sigma^2 | y) \propto \int \sigma^{-n-2} \exp \left\{ -\frac{\sum_i (y_i - \mu)^2}{2\sigma^2} \right\} d\mu.$$

Lemma 12.2 (Sum-of-squares decomposition). *For any $y_1, \dots, y_n$ with mean $\bar{y}$ and $s^2 := \frac{1}{n-1} \sum_i (y_i - \bar{y})^2$,*

$$\sum_{i=1}^n (y_i - \mu)^2 = (n-1)s^2 + n(\bar{y} - \mu)^2.$$

Proof. Write $y_i - \mu = (y_i - \bar{y}) + (\bar{y} - \mu)$ and expand: $\sum_i (y_i - \mu)^2 = \sum_i (y_i - \bar{y})^2 + 2(\bar{y} - \mu) \sum_i (y_i - \bar{y}) + n(\bar{y} - \mu)^2$. The cross term vanishes since $\sum_i (y_i - \bar{y}) = 0$ by definition of the mean. $\square$

Proposition 12.3.

$$\pi(\sigma^2 | y) \propto \sigma^{-(n+1)} \exp \left\{ -\frac{(n-1)s^2}{2\sigma^2} \right\}.$$

Proof. Using Lemma 12.2, the integrand factors into a piece free of μ and a Gaussian integral in μ of variance σ^2/n :

$$\int \sigma^{-n-2} \exp \left\{ -\frac{(n-1)s^2}{2\sigma^2} \right\} \exp \left\{ -\frac{n(\mu - \bar{y})^2}{2\sigma^2} \right\} d\mu = \sigma^{-n-2} \exp \left\{ -\frac{(n-1)s^2}{2\sigma^2} \right\} \cdot \sigma \sqrt{\frac{2\pi}{n}},$$

using $\int_{-\infty}^{\infty} \exp\{-n(\mu - \bar{y})^2/(2\sigma^2)\} d\mu = \sigma \sqrt{2\pi/n}$. Absorbing constants, $\pi(\sigma^2 | y) \propto \sigma^{-n-2} \cdot \sigma \cdot \exp\{-(n-1)s^2/(2\sigma^2)\} = \sigma^{-(n+1)} \exp\{-(n-1)s^2/(2\sigma^2)\}$. $\square$

Corollary 12.4 (Scaled inverse- χ^2 posterior). *Writing $x = \sigma^2$, matching $f(x; \nu, \tau^2) \propto x^{-(\nu/2+1)} \exp\{-\nu\tau^2/(2x)\}$ with $\nu = n-1$, $\tau^2 = s^2$ gives $-(\nu/2+1) = -(n+1)/2$ as a power of $x = \sigma^2$, i.e. a power of $\sigma^{-(n+1)}$ as a power of σ – exactly matching the Proposition above. Hence*

$$\sigma^2 | y \sim \text{Scaled-Inv-}\chi^2(n-1, s^2). \quad (\text{R2})$$

12.4 The Joint Posterior and a Direct (Non-Iterative) Sampling Scheme

Combining (R1) and (R2) via $\pi(\mu, \sigma^2 | y) = \pi(\mu | \sigma^2, y) \pi(\sigma^2 | y)$:

$$\pi(\mu, \sigma^2 | y) \propto \underbrace{\text{N}\left(\bar{y}, \frac{\sigma^2}{n}\right)\text{-kernel in } \mu}_{(\text{R1})} \times \underbrace{\text{Scaled-Inv-}\chi^2(n-1, s^2)\text{-kernel in } \sigma^2}_{(\text{R2})}.$$

Recipe

Because (R2) does not depend on μ , sampling from the joint posterior requires *no iteration* – a simpler, special case of the general Gibbs recipe of Chapter 10:

Step 1. Draw $\sigma^{2(b)} \sim \text{Scaled-Inv-}\chi^2(n-1, s^2)$ from (R2).

Step 2. Draw $\mu^{(b)} | \sigma^{2(b)} \sim \text{N}(\bar{y}, \sigma^{2(b)}/n)$ from (R1), plugging in the just-drawn $\sigma^{2(b)}$.

12.5 Connection to Classical Inference

Proposition 12.5.

$$\frac{(n-1)s^2}{\sigma^2} \Big| y \sim \chi_{n-1}^2.$$

Proof. Immediate from (R2) and Proposition 11.2(a): if $X \sim \text{Scaled-Inv-}\chi^2(\nu, \tau^2)$ then $\nu\tau^2/X \sim \chi_\nu^2$. Here $X = \sigma^2$, $\nu = n-1$, $\tau^2 = s^2$. $\square$

Connecting the Threads

A striking duality. Classical (frequentist) theory proves, by an entirely different route (Cochran's theorem), that for $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ with μ, σ^2 fixed constants and $S^2 = \sum (Y_i - \bar{Y})^2 / (n - 1)$ the sample variance, $(n - 1)S^2 / \sigma^2 \sim \chi_{n-1}^2$ as a *sampling distribution* over repeated draws of the data. Algebraically this is the identical pivotal quantity – but the two statements have completely different meanings. The classical statement fixes σ^2 and treats S^2 (hence the data) as random; the Bayesian statement fixes the observed s^2 (hence the data) and treats σ^2 as the random quantity with a posterior distribution. This is not a coincidence: it is a manifestation of the fact that the flat/Jeffreys reference prior is precisely engineered (Chapter 13) to make Bayesian and frequentist pivotal quantities align numerically, even while meaning something different.

13 Reference Priors via Pivotal Quantities

This chapter gives an independent, more primitive justification for the two reference priors used throughout this book ($\pi(\mu) \propto 1$ for a location parameter, $\pi(\sigma) \propto 1/\sigma$ for a scale parameter) – not via Fisher information (Jeffreys’ route) but via an invariance argument built directly on the structure of the model. That both routes converge on the same priors, by completely different logic, is a strong consistency check on both.

Definition 13.1 (Pivotal quantity). $Q(Y, \theta)$ is *pivotal* if its sampling distribution does not depend on θ .

13.1 Location Families $\Rightarrow$ the Flat Prior

Definition 13.2 (Location family). Y belongs to a location family with location parameter μ if $Y = \mu + U$ for some fixed density g not depending on μ ; equivalently $f(y | \mu) = g(y - \mu)$.

Here $U = Y - \mu$ is manifestly pivotal: $U \sim g$ regardless of μ .

Theorem 13.3. *Requiring that the posterior of $U = Y - \mu$ given data y still equal its fixed sampling distribution g – i.e. $\pi(u | y) = g(u)$, so the model’s invariant structure is preserved after conditioning – forces the (unique, up to constant) prior $\pi(\mu) \propto 1$.*

Proof. By Bayes’ rule, $\pi(\mu | y) \propto f(y | \mu)\pi(\mu) = g(y - \mu)\pi(\mu)$. Change variables from μ to $u = y - \mu$ (so $\mu = y - u$, Jacobian 1): $\pi(u | y) = g(u)\pi(y - u)$. Requiring $\pi(u | y) = g(u)$ for all y forces $\pi(y - u) = 1$ (constant) for all valid $y - u$; since $y - u$ ranges over all of $\mathbb{R}$ as y, u vary, $\pi(\mu)$ must be constant. $\square$

13.2 Scale Families $\Rightarrow$ the $1/\theta$ Prior

Definition 13.4 (Scale family). Y belongs to a scale family with scale parameter $\theta > 0$ if $Y = \theta U$ for fixed density g not depending on θ ; then $U = Y/\theta$ is pivotal, and by change of variables $f(y | \theta) = \frac{1}{\theta}g(y/\theta)$.

Theorem 13.5. *Under the analogous invariance requirement, the unique (up to constant) prior is $\pi(\theta) \propto 1/\theta$.*

Proof. By Bayes’ rule, $\pi(\theta | y) \propto \frac{1}{\theta}g(y/\theta)\pi(\theta)$. Change variables from θ to $u = y/\theta$ (so $\theta = y/u$, $|d\theta/du| = y/u^2$):

$$\pi(u | y) \propto \frac{1}{\theta}g(u)\pi(\theta) \Big|_{\theta=y/u} \cdot \frac{y}{u^2} = \frac{u}{y}g(u)\pi\left(\frac{y}{u}\right) \cdot \frac{y}{u^2} = \frac{g(u)}{u}\pi\left(\frac{y}{u}\right).$$

Trying $\pi(\theta) = 1/\theta$ (up to constant): $\pi(y/u) = u/y$, so $\pi(u | y) \propto \frac{g(u)}{u} \cdot \frac{u}{y} = \frac{g(u)}{y}$. As a function of u (fixed y), the factor $1/y$ is an overall constant that vanishes upon normalizing, so $\pi(u | y) = g(u)$ exactly, independent of y , as required. If instead $\pi(\theta) \neq c/\theta$, the factor $\pi(y/u)/u$ retains genuine dependence on both u and y that cannot be normalized away, so $\pi(u | y)$ would depend on y – violating the requirement. $\square$

Connecting the Threads

Consistency with Jeffreys’ rule. This matches, via a completely different (invariance-based, not information-based) argument, the results of Exercises 9.3–9.4: for $N(\mu, \sigma^2)$ with μ known, σ is a scale parameter and Theorem 13.5 gives $\pi(\sigma) \propto 1/\sigma$ – exactly what the Jeffreys calculation found (there stated as $\pi_J(\sigma^2) \propto 1/\sigma^2$, equivalent under the Jacobian $d\sigma^2/d\sigma = 2\sigma$). Similarly, μ (with σ known) is a location parameter, and Theorem 13.3 gives $\pi(\mu) \propto 1$, matching Exercise 9.3 exactly. Two independent principles – local information geometry (Jeffreys) and global invariance under the natural group of transformations (pivotal quantities) – agree, which is one of the standard justifications for treating these as the canonical reference priors for location and scale parameters.

14 Maximum Likelihood and Maximum a Posteriori Estimation

14.1 The MLE and Its Invariance

Definition 14.1. The likelihood function is $L(\theta; x) := f(x | \theta)$, viewed as a function of θ for fixed observed x . The maximum likelihood estimator is $\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} L(\theta; x)$.

Key Idea

A crucial distinction. $f(x | \theta)$, as a function of x for fixed θ , is a genuine probability density: $\int f(x | \theta) dx = 1$. But $L(\theta; x)$, as a function of θ for fixed x , is generally *not* a density in θ : $\int L(\theta; x) d\theta \neq 1$ in general. This asymmetry is exactly what makes the MLE invariant under reparametrization while the MAP – which maximizes a genuine density $\pi(\theta | x)$ – is not.

Theorem 14.2 (Invariance of the MLE). *Let $\phi = h(\theta)$ be a bijective reparametrization. Then $\hat{\phi}_{\text{MLE}} = h(\hat{\theta}_{\text{MLE}})$.*

Proof. Define $\tilde{L}(\phi; x) := L(h^{-1}(\phi); x)$ – simply relabeling the domain of $L(\cdot; x)$ via the bijection h , with no Jacobian factor, since maximizing a function is a statement about where it is largest, and relabeling the domain by a bijection does not change comparative values. Because h is a bijection, $\max_{\phi} \tilde{L}(\phi; x) = \max_{\theta} L(\theta; x)$, and the maximizer transfers directly: if $\hat{\theta} = \arg \max_{\theta} L(\theta; x)$, then for $\hat{\phi} := h(\hat{\theta})$, $\tilde{L}(\hat{\phi}; x) = L(\hat{\theta}; x) = \max_{\theta} L(\theta; x) = \max_{\phi} \tilde{L}(\phi; x)$, so $\hat{\phi}$ is indeed the maximizer. $\square$

14.2 A Cautionary Example: the Mode Is Not Generally Invariant

Example 14.3 (Mode of a lognormal variable). Let $X \sim N(\mu, \sigma^2)$, with density mode μ . Let $Y = e^X$ (lognormal). Is the mode of Y equal to e^{μ} ?

Solution. By change of variables, $f_Y(y) = f_X(\log y) \cdot \frac{1}{y}$, i.e.

$$f_Y(y) = \frac{1}{y\sigma\sqrt{2\pi}} \exp \left\{ -\frac{(\log y - \mu)^2}{2\sigma^2} \right\}, \quad y > 0.$$

Maximizing $\log f_Y(y) = -\log y - \log(\sigma\sqrt{2\pi}) - \frac{(\log y - \mu)^2}{2\sigma^2}$:

$$\frac{d}{dy} \log f_Y(y) = -\frac{1}{y} - \frac{\log y - \mu}{\sigma^2 y} = 0 \implies 1 + \frac{\log y - \mu}{\sigma^2} = 0 \implies y^* = e^{\mu - \sigma^2} \neq e^{\mu}.$$

The extra $-\sigma^2$ shift comes entirely from the Jacobian factor $1/y$ in the density transformation – a factor not present when merely relabeling a likelihood function’s domain (Theorem 14.2). $\square$

14.3 MAP Estimation and Its Non-Invariance

Definition 14.4. The MAP estimator is the posterior mode: $\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \pi(\theta | x) = \arg \max_{\theta} L(\theta; x)\pi(\theta)$.

Theorem 14.5 (Non-invariance of MAP). *In general $\hat{\phi}_{\text{MAP}} \neq h(\hat{\theta}_{\text{MAP}})$ for a bijective reparametrization $\phi = h(\theta)$; equality holds in general only when h is affine.*

Proof. Because $\pi(\theta | x)$ is a genuine density, transforming to $\phi = h(\theta)$ requires the Jacobian: $\pi(\phi | x) = \pi(\theta(\phi) | x) |d\theta/d\phi|$. Differentiating $\log \pi(\phi | x) = \log \pi(\theta(\phi) | x) + \log |d\theta/d\phi|$ and setting to zero at $\hat{\phi}_{\text{MAP}}$:

$$\left. \frac{\partial \log \pi(\theta | x)}{\partial \theta} \right|_{\theta(\hat{\phi}_{\text{MAP}})} \cdot \frac{d\theta}{d\phi} + \frac{d}{d\phi} \log \left| \frac{d\theta}{d\phi} \right| = 0.$$

If $\hat{\phi}_{\text{MAP}} = h(\hat{\theta}_{\text{MAP}})$ always held, we would need $\partial \log \pi(\theta | x) / \partial \theta = 0$ at $\theta = \hat{\theta}_{\text{MAP}}$ (true) and *also* $\frac{d}{d\phi} \log |d\theta/d\phi| = 0$ at that point – which holds identically for every posterior only if $\log |d\theta/d\phi|$ is constant, i.e. $d\theta/d\phi$ is a constant: precisely the affine case $\phi = a\theta + b$. For a general nonlinear bijection, the extra Jacobian-derivative term is generically nonzero, so the naively transformed point is not the true maximizer. $\square$

14.4 Fully Worked Example: $\text{Exp}(\beta)$ with a Flat Prior

Example 14.6. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\beta)$ with density $f(x | \beta) = \frac{1}{\beta} e^{-x/\beta}$, and take the flat prior $\pi(\beta) \propto 1$ on $\beta > 0$. Find $\hat{\beta}_{\text{MLE}}, \hat{\beta}_{\text{MAP}}$; then reparametrize via $\phi = \log \beta$ and find $\hat{\phi}_{\text{MLE}}, \hat{\phi}_{\text{MAP}}$ directly.

Solution. **MLE.** $\log L(\beta; x) = -n \log \beta - \sum_i x_i / \beta$; differentiating and setting to zero, $\hat{\beta}_{\text{MLE}} = \sum_i x_i / n = \bar{x}$.

MAP in β . Since $\pi(\beta) \propto 1$ is constant, it does not shift the argmax, so $\hat{\beta}_{\text{MAP}} = \hat{\beta}_{\text{MLE}} = \bar{x}$ as well – but this will change once we reparametrize.

$\hat{\phi}_{\text{MLE}}$ *via invariance.* By Theorem 14.2, with $\phi = \log \beta$, $\hat{\phi}_{\text{MLE}} = \log(\sum_i x_i / n)$.

$\hat{\phi}_{\text{MAP}}$ *computed directly.* With $\beta = e^\phi$, $d\beta/d\phi = e^\phi$, the transformed posterior is

$$\pi(\phi | x) \propto \beta^{-n} \exp \left\{ -\frac{\sum_i x_i}{\beta} \right\} \Big|_{\beta=e^\phi} \cdot e^\phi = e^{-(n-1)\phi} \exp \left\{ -\sum_i x_i e^{-\phi} \right\}.$$

Maximizing $\log \pi(\phi | x) = -(n-1)\phi - \sum_i x_i e^{-\phi} + \text{const}$:

$$-(n-1) + \sum_i x_i e^{-\phi} = 0 \implies e^{-\phi} = \frac{n-1}{\sum_i x_i} \implies \hat{\phi}_{\text{MAP}} = \log \left(\frac{\sum_i x_i}{n-1} \right).$$

Conclusion. $\hat{\phi}_{\text{MLE}} = \log(\sum x_i / n) \neq \hat{\phi}_{\text{MAP}} = \log(\sum x_i / (n-1))$: the MAP computed directly in the ϕ -parametrization is a different number from log of the MAP computed in the β -parametrization, an explicit, finite-sample demonstration of Theorem 14.5. (The two agree asymptotically as $n \rightarrow \infty$, since $n/(n-1) \rightarrow 1$: MAP's non-invariance is a finite-sample phenomenon consistent with the general fact that posteriors become approximately invariant and Gaussian as $n \rightarrow \infty$, by the Bernstein–von Mises theorem.) $\square$

Key Idea

A posterior distribution admits more than one natural point summary – the posterior mean $E[\theta | x]$ and the posterior mode $\hat{\theta}_{\text{MAP}}$ are both used in practice, and they respond *differently* to reparametrization. In fact even the posterior mean is not invariant under nonlinear reparametrization (by Jensen's inequality, $E[h(\theta) | x] \neq h(E[\theta | x])$ unless h is affine). No single point estimator is invariant in general; only the full posterior distribution is.

15 Credible Intervals

15.1 Definition and Contrast with Confidence Intervals

Definition 15.1 (Credible interval). A $(1 - \alpha)$ equal-tailed, two-sided credible interval for θ is a pair (L, U) , computed from the data x , such that $\Pr(L < \theta < U \mid x) = 1 - \alpha$, where the probability is taken over the posterior distribution of θ with x fixed at its observed value.

Key Idea

Contrast with the frequentist confidence interval. The Bayesian statement $\Pr(L < \theta < U \mid x) = 1 - \alpha$ and the frequentist statement $\Pr(L < \theta < U) = 1 - \alpha$ look identical but differ in exactly which quantity is random. In the credible interval, θ is the random variable (with its posterior law) and x (hence L, U) is fixed at the observed data – the statement is a direct, post-data claim about where θ lies. In the confidence interval, θ is an unknown but fixed constant, and (L, U) are random functions of not-yet-observed data – the statement is about the *procedure*, over hypothetical repetitions of data collection, not about θ given the one dataset actually observed.

15.2 Equal-Tailed Credible Intervals

The simplest construction takes L, U to be the $\alpha/2$ and $1 - \alpha/2$ quantiles of the posterior,

$$L = \pi^{-1}\left(\frac{\alpha}{2} \mid x\right), \quad U = \pi^{-1}\left(1 - \frac{\alpha}{2} \mid x\right),$$

where $\pi^{-1}(\cdot \mid x)$ is the posterior quantile function. Each tail then carries probability $\alpha/2$, so the middle carries $1 - \alpha$, as required. This is not the only valid construction – a *highest posterior density* (HPD) interval, placed at the region of greatest posterior height rather than at fixed tail probabilities, is generally shorter for skewed posteriors – but equal-tailed intervals are simplest to compute, requiring only the quantile function.

15.3 Worked Example: Binomial Proportion

Proposition 15.2 (Beta–Binomial conjugacy). If $X \mid p \sim \text{Binomial}(n, p)$ and $p \sim \text{Beta}(a, b)$, then $p \mid x \sim \text{Beta}(a + x, b + n - x)$.

Proof. $\pi(p \mid x) \propto L(p; x)\pi(p) \propto p^x(1-p)^{n-x} \cdot p^{a-1}(1-p)^{b-1} = p^{a+x-1}(1-p)^{b+n-x-1}$, the $\text{Beta}(a+x, b+n-x)$ kernel. $\square$

This is exactly Theorem 3.2 from Chapter 3, restated with $x = \sum_i x_i$. The corresponding 95% equal-tailed credible interval is

$$\left[F_{\text{Beta}(a+x, b+n-x)}^{-1}(0.025), F_{\text{Beta}(a+x, b+n-x)}^{-1}(0.975) \right],$$

evaluated numerically, since the Beta CDF has no elementary closed form for general parameters – this is the precise construction anticipated at the end of Chapter 2’s posterior-summary discussion.

15.4 Worked Example: Normal Mean, Known Variance

Proposition 15.3. Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, σ known, $\pi(\mu) \propto 1$. Then $\mu \mid x \sim N(\bar{x}, \sigma^2/n)$.

Proof. $\pi(\mu \mid x) \propto \exp\{-\sum_i (x_i - \mu)^2 / (2\sigma^2)\}$. By Lemma 12.2 (with x_i in place of y_i), $\sum_i (x_i - \mu)^2 = \sum_i (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$; the first term is μ -free and absorbs into the proportionality constant, leaving $\pi(\mu \mid x) \propto \exp\{-n(\mu - \bar{x})^2 / (2\sigma^2)\}$, the $N(\bar{x}, \sigma^2/n)$ kernel. $\square$

This is a special case of (R1) (no nuisance parameter to worry about, since σ is known rather than estimated). The resulting 95% equal-tailed credible interval, using the standard normal quantile $z_{0.975} = 1.96$, is

$$\left[\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right],$$

numerically identical to the classical confidence interval for μ – but, as emphasized above, with a different meaning: it is a direct probability-0.95 statement about μ given the observed $\bar{x}$, made possible precisely because the flat reference prior was used.

Consolidated Summary and Master Reference Tables

This closing chapter gathers, in one place, every conjugate model, every Jeffreys prior, and every summary formula developed across Parts I–V, so that the book can be used as a working reference after a first reading.

Master Table 1: Conjugate Models

	Beta–Bernoulli	Poisson–Gamma	Normal–Normal (σ^2 known)
Likelihood	$X_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta)$	$X_i \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$	$Y_i \stackrel{\text{iid}}{\sim} \text{N}(\theta, \sigma^2)$
Prior	$\text{Beta}(\alpha, \beta)$	$\text{Gamma}(a, b)$ (implicit via Jeffreys)	$\text{N}(\mu_0, \tau_0^2)$
Posterior	$\text{Beta}(\alpha + \sum x_i, \beta + n - \sum x_i)$	$\text{Gamma}(a + \sum x_i, b + n)$	$\text{N}(\mu_1, \tau_1^2)$
Posterior mean	$\frac{\alpha + \sum x_i}{\alpha + \beta + n}$	$\frac{a + \sum x_i}{b + n}$	precision-weighted average of $\mu_0, \bar{y}$
Sufficient stat.	$\sum_i x_i$	$\sum_i x_i$	$\bar{y}$
Consistency	posterior var. $\rightarrow 0$, mean $\rightarrow \bar{x}$	mean $\rightarrow \bar{x}$	$\tau_1^2 \rightarrow 0$, mean $\rightarrow \bar{y}$

Master Table 2: Jeffreys Priors

Model	$I(\theta)$	$\pi_J(\theta)$	Family/type
$\text{Poisson}(\lambda)$	$1/\lambda$	$\lambda^{-1/2}$	scale-like; improper, proper posterior
$\text{Binomial}(n, p)$	$n/[p(1-p)]$	$\text{Beta}(\frac{1}{2}, \frac{1}{2})$	proper (arcsine distribution)
$\text{N}(\mu, \sigma^2)$, σ^2 known	$1/\sigma^2$	$\propto 1$	location; flat
$\text{N}(\mu, \sigma^2)$, μ known	$1/(2\sigma^4)$	$\propto 1/\sigma^2$	scale; improper
Poisson regression	$X^\top W X$	$\propto \sqrt{\det(X^\top W X)}$	matrix generalization

Master Table 3: The Book’s Central Recurring Phenomena

Phenomenon	Where it appears
Shrinkage: posterior mean is a weighted average of prior guess and data	Laplace’s rule of succession (Ch. 2); general Beta–Bernoulli (Prop. 3.4); Normal–Normal precision-weighting (Thm. 4.1, 4.2)
Posterior consistency: mean $\rightarrow$ MLE, variance $\rightarrow 0$ as $n \rightarrow \infty$	Beta–Bernoulli (Ch. 3); Normal–Normal (Ch. 4); Bernstein–von Mises intuition (Ch. 14)
Completing the square as the master algebraic tool	Normal–Normal single obs. (Thm. 4.1), n obs. (Thm. 4.2), marginal y (Thm. 4.3), full (μ, σ^2) model (Ch. 12)
Two independent routes to the same reference prior	Jeffreys/Fisher information (Ch. 6–9) versus pivotal quantities (Ch. 13)
Invariance under reparametrization: who has it, who doesn’t	MLE invariant (Thm. 14.2); Jeffreys prior invariant (Thm. 6.4); posterior mode/MAP <i>not</i> invariant (Thm. 14.5)
Marginalizing nuisance parameters by simulation	Monte Carlo marginalization (Prop. 10.3); Gibbs sampling (Ch. 10); direct two-step sampler for (μ, σ^2) (Ch. 12)

This closes the arc that opened in Chapter 1: Step 1 (build a joint model), Step 2 (condition to get a posterior), and – now fully realized – Step 3 (summarize and use that posterior, whether through a point estimate, a credible interval, or a predictive distribution). Every later chapter was, in the end, one more worked instance of these same three steps.